National Comprehensive Cancer Network®

NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)

Kaposi Sarcoma

Version 2.2026 — September 16, 2025

NCCN recognizes the importance of clinical trials and encourages participation when applicable and available. Trials should be designed to maximize inclusiveness and broad representative enrollment.

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NCCN Guidelines Panel Disclosures

ξ Bone marrow transplantation
ϖ Dermatology
Ω Gynecologic oncology
‡ Hematology/Hematology oncology
Φ Infectious diseases
Þ Internal medicine
† Medical oncology
≠ Pathology
¥ Patient advocacy
Σ Pharmacology
§ Radiotherapy/Radiation oncology
¶ Surgery/Surgical oncology
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NCCN Kaposi Sarcoma Panel Members
Summary of Guidelines Updates

- Diagnosis and Workup (KS-1)
- Limited Cutaneous Disease (KS-2)
- Advanced Disease (KS-3)
- Surveillance (KS-4)
- KSHV-Associated Inflammatory Cytokine Syndrome (KICS) (KICS-1)
- Staging Classification and Response Definitions for KS (KS-A)
- Principles and Goals of Therapy (KS-B)
- Principles of Immune Reconstitution Inflammatory Syndrome (IRIS) (KS-C)
- Local Therapy (KS-D)
- Principles of Radiation Therapy (KS-E)
- Systemic Therapy (KS-F)

- Abbreviations (ABBR-1)

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NCCN Categories of Evidence and Consensus: All recommendations are category 2A unless otherwise indicated.

See NCCN Categories of Evidence and Consensus.

NCCN Categories of Preference: All recommendations are considered appropriate.

See NCCN Categories of Preference.

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Updates in Version 2.2026 of the NCCN Guidelines for Kaposi Sarcoma from Version 1.2026 include:

MS-1
• The discussion section has been updated to reflect the changes in the algorithm.

Updates in Version 1.2026 of the NCCN Guidelines for Kaposi Sarcoma from Version 2.2025 include:

Global
• References were updated throughout the Guidelines.

KS-1
• Diagnosis, Useful in Certain Circumstances, last bullet added: Encourage cytology and flow cytometry of effusion fluid if present to evaluate for coexisting primary effusion lymphoma (PEL).
• Workup
  ▸ Essential, 1st bullet, 1st sub-bullet revised: Including history of additional immunosuppressive disease or *immunomodulatory* therapy *(eg, transplant; local or systemic corticosteroids)*~~, such as transplant/,local or systemic glucocorticoids~~.
  ▸ Useful in Selected Cases
    ◊ Bullet 3 added: Stool hemoccult and/or chest x-ray as clinically indicated (eg, for advanced disease).
    ◊ Bullets removed:
      – In the setting of advanced disease cutaneous, oral, visceral, or nodal involvement: stool hemoccult.
      – In the setting of advanced cutaneous, oral, visceral, or nodal disease, or any pulmonary symptoms: chest x-ray.
    ◊ 6th bullet revised: If concerns for coexisting KSHV-associated inflammatory cytokine syndrome (KICS), MCD, or KSHV+ lymphoma, or other KSHV-associated disease: *CT scan or* FDG-PET/CT scan and/or lab workup.
• Header revised: KS ~~STAGE~~ STATE (Also for KS-2 and KS-3)
  ▸ KS STATE
    ◊ Qualifiers removed (cutaneous, oral, visceral, or nodal) from Advanced disease pathway and added to footnote e. (Also for KS-2 and KS-3)
    ◊ Pathway added: KSHV-Associated Inflammatory Cytokine Syndrome (KICS) (KICS-1)
• Footnote a added: Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See NCCN Guidelines for Distress Management (DIS-A).
• Footnote c revised: Imaging should be directed by symptoms or findings concerning for visceral or bone involvement as well as coexisting KICS, MCD, or KSHV+ lymphoma and can include whole body FDG-PET/CT. See ~~NCCN Guidelines for B-Cell Lymphomas (CD-1)~~ *NCCN Guidelines for Castleman Disease*. (Also for KS-4)
• Footnote e added: Advanced disease refers to disease that is not limited to cutaneous disease (eg, advanced cutaneous, oral, visceral, or nodal disease). See Staging and Classification for KS (KS-A 1 of 2) for definitions of T0 and T1. (Also for KS-2 and KS-3)
• Footnote f added: Patients with KICS present with symptoms such as fever, fatigue, edema, cachexia, gastrointestinal or respiratory symptoms, pain, neuropathy, and/or an altered mental state.

KS-2
• Footnote r added: See Response Definitions for KS (KS-A 2 of 2). (Also for KS-3)

Continued
UPDATES

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Updates in Version 1.2026 of the NCCN Guidelines for Kaposi Sarcoma from Version 2.2025 include:

KS-4
- Bullet 1 added: Patients should be educated on the signs of disease progression and encouraged to promptly report any concerns.
- 2nd bullet, 1st sub-bullet revised: *Periodic, life-long follow-up should be patient-centered with timing based* on response to therapy and, if applicable, *presence of immunosuppressive therapy or condition including* degree of HIV viremia and immune reconstitution.
- Sub-sub-bullet 1 added: Dermatology, medical oncology, hematology, and primary HIV care should be included as appropriate.
- Bullet 4, sub-bullet 2 added: Stool hemoccult and/or chest x-ray as clinically indicated (eg, for advanced disease).
- Bullet 4, sub-sub bullet 1 added: When effusions are present, encourage cytology and flow cytometry to evaluate for PEL.
- Bullet 4, sub-bullet removed: In the setting of advanced cutaneous, oral, visceral, or nodal involvement: stool hemoccult.
- Bullet 4, sub-bullet removed: In the setting of advanced cutaneous, oral, visceral, or nodal disease: chest x-ray.
- Footnote v added: For additional considerations in the post-treatment setting, see the NCCN Guidelines for Survivorship.

KICS-1
- New Algorithm: KSHV-Associated Inflammatory Cytokine Syndrome (KICS)

KS-A (2 of 2)
- Footnote b added: Although this response criteria was developed for HIV-associated KS, it is applied to all forms of KS in clinical trials.

KS-B
- Principles of Therapy
  - 2nd bullet revised: Optimization of immune function and avoidance of additional immunosuppression are critical to prevention of additional KS lesions and maintenance of response to therapy. For HIV-related KS, reconstitution of immune function and maintenance of viral suppression are important. Also, it is important to work with an HIV specialist to optimize suppression of HIV and reconstitution of immune function with ART. *In patients with KS who require immunosuppression, collaboration with a pharmacist to discuss treatment options is recommended.*
- Goals of Therapy
  - 2nd bullet revised: Patients with symptomatic or cosmetically bothersome disease should use minimally invasive and the least toxic therapy to control disease *(ie, local therapy, RT, limited systemic therapy)*. When necessary, a limited number of cycles of systemic therapy (eg, 3–6) may be sufficient for ~~those~~ *PWH* initiating or re-initiating ART.
  - 3rd bullet revised: Patients with advanced symptomatic ~~cutaneous, visceral, nodal, or oral~~ disease should be treated with systemic therapy with the goal of reducing or reversing symptoms, lymphedema, or threat to organ function. Complete remissions are rare.

KS-C
- Management
  - 2nd bullet, 1st sub-bullet revised: ~~Consider thalidomide~~ *Corticosteroid-sparing treatment such as thalidomide can be considered* as an active agent against both KS and ~~corticosteroid-refractory~~ IRIS.

Continued
UPDATES

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Updates in Version 1.2026 of the NCCN Guidelines for Kaposi Sarcoma from Version 2.2025 include:

KS-F (1 of 4)
- First-line systemic therapy options revised:
  - *For non-transplant KS*
    - ◊ Preferred
      - – Liposomal Doxorubicin
    - ◊ Other recommended
      - – Paclitaxel
  - *For transplant KS*
    - ◊ Preferred
      - – Sirolimus
    - ◊ Other recommended
      - – Liposomal Doxorubicin
      - – Paclitaxel
- Subsequent systemic therapy options for relapsed/refractory therapy revised (Also for KS-F 2):
  - Nivolumab and Pembrolizumab changed from useful in certain circumstances.
- Useful in certain circumstances revised:
  - Bortezomib, Gemcitabine, Lenalidomide, and Vinorelbine changed from preferred.

KS-F (2 of 4)
- First-Line Systemic Therapy Dosing, Other recommended regimen, sub-bullet removed: Premedication with dexamethasone may not be needed; if used, the dose should be minimized and tailored to patient needs.

KS-F (3 of 4)
- Footnote b added: Corticosteroid pre-medication as an anti-emetic, or to prevent infusion/hypersensitivity reaction (except in cases of prior reaction) should generally be avoided.
- Footnote c added: An FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN guidelines.

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DIAGNOSIS

ESSENTIAL:
- Review of adequate slides from paraffin block representative of the tumor by a pathologist with expertise in the diagnosis of Kaposi sarcoma (KS)
- Rebiopsy if non-diagnostic
- Histopathology review of adequate biopsy (ie, skin punch, incisional, excisional)
- Adequate immunophenotyping to establish diagnosis
- Immunohistochemistry (IHC) panel: Kaposi sarcoma-associated herpesvirus (KSHV; human herpesvirus 8 [HHV8]), LANA-1

USEFUL IN CERTAIN CIRCUMSTANCES:
- IHC: CD31 and CD34 if unclear whether the tumor has a vascular origin
- Encourage additional biopsy of nodal or visceral sites if a coexisting disorder is suspected (ie, infection, lymphoma, multicentric Castleman disease [MCD])
- Encourage cytology and flow cytometry of effusion fluid if present to evaluate for coexisting primary effusion lymphoma (PEL)

WORKUP[a]

ESSENTIAL:
- History and physical examination
  - Including history of additional immunosuppressive disease or immunomodulatory therapy (eg, transplant; local or systemic corticosteroids)
  - Including complete skin, oral, and lymph node examinations, and evaluation of edema
- Complete blood count (CBC), differential, and comprehensive metabolic panel
- Human immunodeficiency virus (HIV) screening and/or diagnostic testing[b]
- Photography of oral, conjunctival, and cutaneous lesions (with reference unit of measure in the picture) for evaluation and monitoring of extent of disease

USEFUL IN SELECTED CASES:
- In patients of childbearing potential if chemotherapy or radiation therapy (RT) planned: pregnancy testing
- Evaluation[b] for suspected opportunistic infections (OIs)
- Stool hemoccult and/or chest x-ray as clinically indicated (eg, for advanced disease)
- If unexplained pulmonary symptoms or abnormalities on chest x-ray: chest CT with contrast[c] and bronchoscopy
- If gastrointestinal (GI) symptoms or positive hemoccult: abdomen/pelvis CT with contrast[c] or MRI with and without contrast[c] and esophagogastroduodenoscopy (EGD)/colonoscopy
- If concerns for coexisting KSHV-associated inflammatory cytokine syndrome (KICS), MCD, or KSHV+ lymphoma, or other KSHV-associated disease: CT scan or FDG-PET/CT scan[c] and/or lab workup[d]
- If anthracycline planned or suspected pericardial effusion: transthoracic echocardiogram
- For people with HIV (PWH), begin discussions regarding the possible need to modify ART due to DDIs and the need to involve an HIV specialist in care decisions. See Principles of Systemic Therapy and Drug-Drug Interactions (HIV-B) in the NCCN Guidelines for Patients with HIV.

KS STATE[e]

- Limited cutaneous disease → First-Line Therapy (KS-2)
- Advanced disease → First-Line Therapy (KS-3)
- KSHV-Associated Inflammatory Cytokine Syndrome (KICS)[f] (KICS-1)

[a] Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See NCCN Guidelines for Distress Management (DIS-A).
[b] All patients who are HIV seropositive should have recent T-cell subsets, including quantitative CD4+ T-cell count and HIV viral load to assess immune function and HIV control (Discussion). Involvement of an HIV specialist to evaluate for coexisting OI is appropriate, especially with advanced immunosuppression.
[c] Imaging should be directed by symptoms or findings concerning for visceral or bone involvement as well as coexisting KICS, MCD, or KSHV+ lymphoma and can include whole body FDG-PET/CT. See NCCN Guidelines for Castleman Disease.
[d] Useful in patients with clinical features (ie, fever, dyspnea, effusions) concerning for KICS or KSHV-associated MCD: C-reactive protein, KSHV serum viral load, serum protein electrophoresis (SPEP), interleukin (IL)-6, or IL-10.
[e] Advanced disease refers to T1 disease, extensive T0 cutaneous disease, or nodal disease. See Staging Classification for KS (KS-A 1 of 2) for definitions of T0 and T1.
[f] Patients with KICS present with symptoms such as fever, fatigue, edema, cachexia, gastrointestinal or respiratory symptoms, pain, neuropathy, and/or an altered mental state.

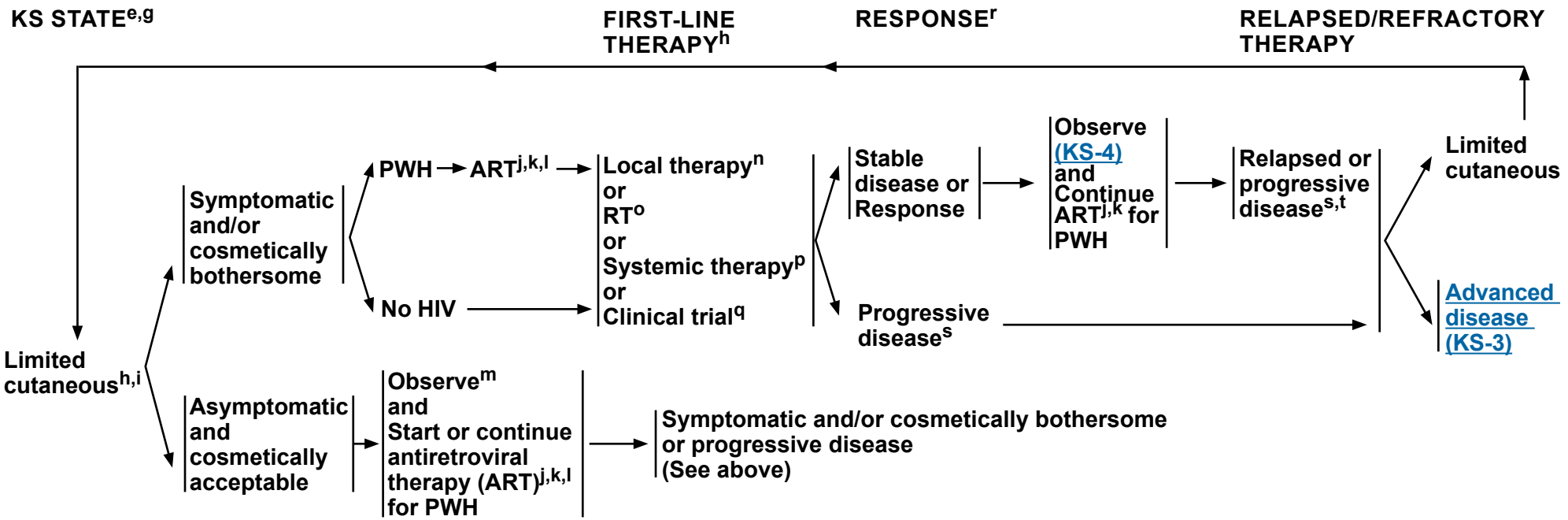
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[e] Advanced disease refers to T1 disease, extensive T0 cutaneous disease, or nodal disease. See Staging Classification for KS (KS-A 1 of 2) for definitions of T0 and T1.

[g] Oncology and HIV clinicians, along with both an oncology pharmacist and HIV pharmacist, if available, should review proposed cancer therapy, supportive care medications, and ART for possible drug-drug interactions (DDIs) and overlapping toxicities prior to initiation. Co-management by an oncologist and an HIV clinician is recommended for the duration of therapy. See NCCN Guidelines for Cancer in People with HIV.

[h] Principles and Goals of Therapy (KS-B).

[i] There is no agreed upon definition for limited cutaneous disease, but some trials have used inclusion criteria such as 5 index lesions or <5% body surface area involvement as the threshold. Clinical judgment is required.

[j] Initiation of ART may result in immune reconstitution inflammatory syndrome (IRIS) within 3–6 months; IRIS is characterized by marked lesional swelling, increased tenderness, and peripheral edema. Therefore, systemic KS-directed therapy should be initiated as soon as possible in all PWH who have symptomatic KS (eg, edema, oral KS, or other manifestation that interfere with normal functions or activities). In patients with limited cutaneous disease, KS-associated IRIS is an indication for urgent initiation of systemic KS therapy; signs of past IRIS may also be an indication for earlier initiation of systemic KS therapy. ART should not be delayed or discontinued unless life-threatening IRIS develops. Reconstitution of immune function is important for obtaining and maintaining control or remission of KS. See Principles of Immune Reconstitution Inflammatory Syndrome (IRIS) (KS-C).

[k] Glucocorticoids in any formulation should be avoided due to their association with KS progression. However, in cases of life-threatening conditions, their use may be considered.

[l] Patients who are only on ART should be reassessed within 4 weeks, particularly to monitor for KS-IRIS.

[m] Rapid progression during observation may be an indication for initiation of therapy.

[n] Local Therapy (KS-D).

[o] Principles of Radiation Therapy (KS-E).

[p] Systemic Therapy (KS-F).

[q] clinicaltrials.gov.

[r] See Response Definitions for KS (KS-A 2 of 2).

[s] If progressive, relapsed, or refractory disease, consider biopsy of lesions since they may be areas of post-inflammatory pigment and/or mimickers of KS. If KS, evaluate for inadequate HIV control as a contributing factor to inadequate KS control and address possible change in ART in conjunction with an HIV specialist. See NCCN Guidelines for Cancer in People with HIV.

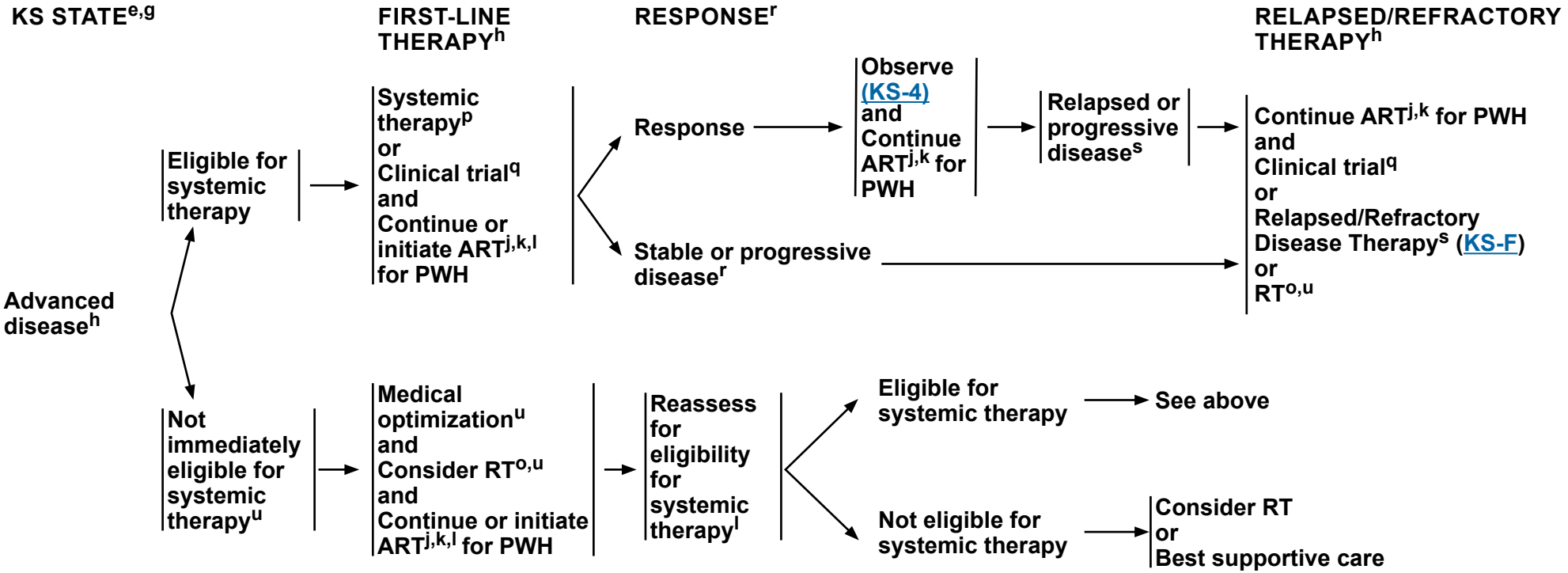
[t] If after initial response to therapy, KS relapses or progresses, repeat use of previously effective therapy may be considered, particularly if response was durable.

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e Advanced disease refers to T1 disease, extensive T0 cutaneous disease, or nodal disease. See Staging Classification for KS (KS-A 1 of 2) for definitions of T0 and T1.

g Oncology and HIV clinicians, along with both an oncology pharmacist and HIV pharmacist, if available, should review proposed cancer therapy, supportive care medications, and ART for possible DDIs and overlapping toxicities prior to initiation. Co-management by oncologist and HIV clinician is recommended for the duration of therapy. See NCCN Guidelines for Cancer in People with HIV.

h Principles and Goals of Therapy (KS-B).

j Initiation of ART may result in IRIS within 3–6 months; IRIS is characterized by marked lesional swelling, increased tenderness, and peripheral edema. Therefore, systemic KS-directed therapy should be initiated as soon as possible in all PWH who have symptomatic KS (eg, edema, oral KS, or other manifestation that interfere with normal functions or activities). In patients with limited cutaneous disease, KS-associated IRIS is an indication for urgent initiation of systemic KS therapy; signs of past IRIS may also be an indication for earlier initiation of systemic KS therapy. ART should not be delayed or discontinued unless life-threatening IRIS develops. Reconstitution of immune function is important for obtaining and maintaining control or remission of KS. See Principles of Immune Reconstitution Inflammatory Syndrome (IRIS) (KS-C).

k Glucocorticoids in any formulation should be avoided due to their association with KS progression. However, in cases of life-threatening conditions, their use may be considered.

l Patients who are only on ART should be reassessed within 4 weeks, particularly to monitor for KS-IRIS.

o Principles of Radiation Therapy (KS-E).

p Systemic Therapy (KS-F).

q clinicaltrials.gov.

r See Response Definitions for KS (KS-A 2 of 2).

s If progressive, relapsed, or refractory disease, consider biopsy of lesions since they may be areas of post-inflammatory pigment and/or mimickers of KS. If KS, evaluate for inadequate HIV control as a contributing factor to inadequate KS control and address possible change in ART in conjunction with an HIV specialist. See NCCN Guidelines for Cancer in People with HIV.

u Systemic therapy is preferred over RT as first-line therapy and relapsed/refractory therapy for disseminated disease whenever systemic therapy is feasible. Although uncommon, performance status and medical comorbidities may preclude the use of upfront chemotherapy in some patients. Reversible medical issues should be addressed (eg, active infection requiring antibiotic or antiviral treatment; high-level HIV viremia requiring initiation/modification of ART) so that systemic therapy can be delivered as quickly as possible. If poor performance status is because of KS, systemic therapy should be considered because it may lead to an improvement in performance status.

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SURVEILLANCE[v]

- **Patients should be educated on the signs of disease progression and encouraged to promptly report any concerns**
- **For patients not requiring active therapy and with no signs of progression**
  - **Periodic, life-long follow-up should be patient-centered with timing based on response to therapy and, if applicable, presence of immunosuppressive therapy or condition including degree of HIV viremia and immune reconstitution**
    - ◊ **Dermatology, medical oncology, hematology, and primary HIV care should be included as appropriate**
    - ◊ **Follow-up should include the following:**
      - **History and physical examination**
        - **Including history of additional immunosuppression such as transplant/glucocorticoids**
        - **Including complete skin and oral examinations, and evaluation and monitoring of edema**
      - **CBC, differential, comprehensive metabolic panel**
      - **PWH**
        - **T-cell subsets (CD4+ T-cell count) and HIV viral load**
        - **Assess ART compliance**
- **Photography of oral, conjunctival, and cutaneous lesions (with reference unit of measure in the picture) for evaluation of extent of disease if change in disease is noted**
- **If signs and symptoms concerning for visceral involvement or prior to new therapy if progression/refractory disease or if change in disease is noted, the following are indicated depending on the clinical circumstances[c]:**
  - **Evaluation for suspected OIs**
  - **Stool hemoccult and/or chest x-ray as clinically indicated (eg, for advanced disease)**
  - **If unexplained pulmonary symptoms or abnormalities on chest x-ray: chest CT with contrast[c] and bronchoscopy**
  - **If GI symptoms or positive hemoccult: abdomen/pelvis CT[c] with contrast or MRI with and without contrast[c] and EGD/colonoscopy**
  - **If concerns for coexisting KICS, MCD, or KSHV+ lymphoma, or other KSHV-associated disease: FDG-PET/CT[c] and/or lab workup[d]**
    - ◊ **When effusions are present, encourage cytology and flow cytometry to evaluate for PEL.**
  - **The diagnosis of KICS ideally requires excisional biopsy of lymphadenopathy to exclude MCD, if feasible**
- **As KSHV is not eradicated with treatment of KS, the risk for future KS persists even after complete remission.**
- **For PWH, optimization and monitoring of HIV control and immune function is important to minimize this risk. This risk depends on immune function and generally decreases with immune reconstitution. However, KS can persist, relapse, or present even in the setting of normal values of T-cell subsets. Less frequent (every 6–12 mo) oncology monitoring may be appropriate for selected patients with undetectable HIV viral loads, normal T-cell subsets, and stable KS for 2 or more years as long as the patient has regular follow-up with an HIV provider.**

[c] Imaging should be directed by symptoms or findings concerning for visceral or bone involvement as well as coexisting KICS, MCD, or KSHV+ lymphoma and can include whole body FDG-PET/CT. See NCCN Guidelines for Castleman Disease.
[d] Useful in patients with clinical features (ie, fever, dyspnea, effusions) concerning for KICS or KSHV-associated MCD: C-reactive protein, KSHV serum viral load, SPEP, IL-6, or IL-10.
[v] For additional considerations in the post-treatment setting, see NCCN Guidelines for Survivorship.

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NCCN Guidelines Version 2.2026
KSHV-Associated Inflammatory Cytokine Syndrome (KICS)

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Discussion

ADDITIONAL WORKUP[a,1-3]

ESSENTIAL:

- Lymph node biopsy with KSHV specific evaluations such as LANA-1 (see KS-1 and NCCN Guidelines for Castleman Disease)
- Cytology of effusions and flow cytometry to rule-out PEL, including KSHV-specific evaluations such as LANA

USEFUL IN CERTAIN CIRCUMSTANCES:

- Bone marrow biopsy if MCD (TAFRO syndrome) or HLH or lymphoma is suspected
- CT scan or FDG-PET/CT scan and lab workup if lymphoma suspected (see KS-1)

TREATMENT[b,c]

- Systemic therapy
  - Preferred
    - ◊ Liposomal Doxorubicin + Rituximab[d]
  - Other recommended
    - ◊ Alternative KS-directed systemic therapies + Rituximab[d] may be considered (see KS-F)
  - In mild cases, KS-directed therapy alone may be adequate; Rituximab[d] may be added later if inadequate response
  - For transplant recipients, the Sirolimus component of immunosuppression should be maximized and other components should be minimized
- Early consultation of KS experts is encouraged

MONITORING

- Surveillance (see KS-4)
- Additional monitoring for improvement/normalization of symptoms (eg, lymphadenopathy, splenomegaly, cytopenias, inflammatory markers) to determine duration of KICS therapy
  - CBC, differential
  - CMP and CRP
  - KSHV serum viral load, IL-6, IL-10 (if feasible)
  - Imaging as indicated (eg, as follow-up for lymphadenopathy/organomegaly)
- Consider repeating cytology of effusions and biopsy of lymphadenopathy in refractory/progressive disease

Refractory/Progressive Disease → Continue current KS-directed therapy and consider adding anti-IL-6-directed therapy (Tocilizumab or Siltuximab[e])

Inflammatory syndrome and lymphadenopathy/organomegaly improving →

- Rituximab can be withheld while KS-directed therapy is continued for 3 months or longer as needed for adequate/optimal KS response[f]
- If inflammatory symptoms/lab markers flare after Rituximab is stopped, Rituximab should be resumed

Footnotes

[a] See Diagnosis and Workup on KS-1. Differential diagnoses for KICS include MCD including TAFRO syndrome (see NCCN Guidelines for Castleman Disease); KS-IRIS (see KS-C); lymphoma, especially PEL; sepsis; tuberculosis (TB)/mycobacterial infection; hemophagocytic lymphohistiocytosis (HLH); and immunotherapy-related cytokine release syndrome.

[b] Steroids should generally be avoided because they may exacerbate symptoms of KICS.

[c] An FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN guidelines.

[d] Rituximab and hyaluronidase human injection for subcutaneous use may be substituted for rituximab after patients have received the first full dose of rituximab by intravenous infusion.

[e] Tocilizumab blocks the IL-6 receptor, but not the portion bound by viral IL-6. Siltuximab binds to human IL-6, preventing it from activating the IL-6 receptor.

[f] Because flares of KS may occur due to prolonged Rituximab effect, KS-directed therapy should be continued for at least 3 months after cessation of rituximab. Systemic therapy should be continued until there is an adequate KS response.

References

[1] Polizzotto MN, Uldrick TS, Wyvill KM, et al. Clin Infect Dis 2016;62:730-738.
[2] Gonçalves PH, Uldrick TS, Yarchoan R. AIDS 2017;31:1903-1916.
[3] Karass M, Grossniklaus E, Seoud T, et al. Oncologist 2017;22:623-625.

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STAGING CLASSIFICATION FOR KS[a]

	Good risk (all of the following)	Poor risk (any of the following)
Tumor, T	**T0: Confined to skin and/or lymph nodes and/or minimal oral disease (non-nodular KS confined to palate)**	**T1: Tumor-associated edema or ulceration Extensive oral KS Gastrointestinal KS KS in organs other than lymph nodes**
Immune system, I[1]	**I0: CD4+ T-cell count ≥150/μL**	**I1: CD4+ T-cell count <150/μL**
Systemic disease, S	**S0: No history of opportunistic infection or thrush No "B" symptoms[2] Karnofsky Performance Status ≥70**	**S1: History of opportunistic infection and/or thrush "B" symptoms present Karnofsky Performance Status <70 Other HIV-related illness (eg, neurologic disease, lymphoma)**

[1] I stage has less prognostic value than T or S stages in patients on ART therapy.
[2] "B" symptoms are unexplained fever, night sweats, >10% involuntary weight loss, or diarrhea persisting >2 weeks.

[a] Adapted from Krown SE, Metroka C, Wernz JC. Kaposi's sarcoma in the acquired immune deficiency syndrome: a proposal for uniform evaluation, response, and staging criteria. AIDS Clinical Trials Group Oncology Committee. J Clin Oncol 1989;7:1201-1207.

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RESPONSE DEFINITIONS FOR KS[a,b]

Complete response (CR)	**The absence of any detectable residual disease, including tumor-associated (local) edema, persisting for at least 4 weeks. Patients known to have had visceral disease should have restaging with appropriate endoscopic or radiographic procedures relevant to sites involved at baseline.**
Partial response (PR)	**No new mucocutaneous lesions, visceral sites of involvement, or the appearance or worsening of tumor-associated edema or effusions; AND** **▸ A 50% or greater decrease in the number of all previous existing lesions lasting for at least 4 weeks; OR** **▸ Complete flattening of at least 50% of all previously raised lesions (ie, 50% of all previously nodular or plaque-like lesions become macules); OR** **▸ A 50% decrease in the sum of the products of the largest perpendicular diameters of at least 5 measurable lesions.** **NOTE: When there is residual tumor-associated edema or effusion, but disease otherwise meets criteria for complete response, response should be classified as "partial."**
Stable disease (SD)	**Any response that does not meet the criteria for progressive disease or PR.**
Progressive disease (PD)	**An increase of ≥25% in the size of pre-existing lesions and/or the appearance of new lesions or sites of disease and/or a change in the character of the skin or oral lesions from macular to plaque-like or nodular of ≥25%. If new or increasing tumor-associated edema or effusion develop, disease is considered to be progressive.**

[a] Adapted from Krown SE, Metroka C, Wernz JC. Kaposi's sarcoma in the acquired immune deficiency syndrome: a proposal for uniform evaluation, response, and staging criteria. AIDS Clinical Trials Group Oncology Committee. J Clin Oncol 1989;7:1201-1207.
[b] Although this response criteria was developed for HIV-related KS, it is applied to all forms of KS in clinical trials.

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PRINCIPLES AND GOALS OF THERAPY

Principles of Therapy:

- **Individual KS lesions may be distinct clones that arise due to the common risk factors of immunosuppression and persistent KSHV infection as opposed to metastases. Treatment of existing disease therefore may not prevent occurrence of future lesions.**
- **Optimization of immune function and avoidance of additional immunosuppression are critical to prevention of additional KS lesions and maintenance of response to therapy. For HIV-related KS, reconstitution of immune function and maintenance of viral suppression are important. Also, it is important to work with an HIV specialist to optimize suppression of HIV and reconstitution of immune function with ART. In patients with KS who require immunosuppression, collaboration with a pharmacist to discuss treatment options is recommended.**
  - **Important examples of iatrogenic immunosuppression, which may promote KS, include not only systemic but local glucocorticoids (ie, inhaled, topical, intra-articular). Note that KS may flare in a remote location from the site of local glucocorticoids.**
    - **Glucocorticoids in any formulation should be avoided due to their association with KS progression.[1-3] However, in cases of life-threatening conditions, their use may be considered. Proceed with caution if using glucocorticoids or consult an HIV or KS specialist.**
  - **Patients requiring rituximab for treatment of non-Hodgkin lymphoma (NHL) with coexisting KS or MCD may develop flares of KS or incident KS. This may be mitigated by use of concurrent chemotherapy active against both KS and disease for which rituximab is prescribed (ie, doxorubicin).**
- **Persons with HIV-related KS, especially those with advanced immunosuppression, are at increased risk of OIs, marrow suppression with neutropenic fever, or thrombocytopenic bleeding and should be monitored closely. It is important to collaborate with an HIV specialist to ensure adequate OI prophylaxis appropriate to CD4+ T-cell count (which may temporarily decrease with cytotoxic chemotherapy). Growth factor support may be needed to facilitate systemic therapy.**
- **Lymphedema and soft tissue infections: KS is often complicated by lymphedema with increased risk of cellulitis and deep tissue infections in affected limbs. Risk of severe lymphedema and delayed wound healing may be increased after radiation. Refer to a lymphedema specialist. In the setting of advanced cutaneous disease, radiation should be reserved for circumstances when systemic therapy is not feasible with the goal of palliation or short-term disease management until systemic therapy may be delivered. Note that treatment responses may be delayed in the context of significant lymphedema.**

Goals of Therapy:

- **PWH with limited cutaneous disease that is asymptomatic and cosmetically acceptable may be observed while starting or continuing ART with optimization of immune function and HIV viral suppression as above. Remissions or stable disease may occur with ART and optimization of immune function and HIV viral suppression alone.**
- **Patients with symptomatic or cosmetically bothersome disease should use minimally invasive and the least toxic therapy to control disease (ie, local therapy, RT, limited systemic therapy). When necessary, a limited number of cycles of systemic therapy (eg, 3–6) may be sufficient for PWH initiating or re-initiating ART.**
- **Patients with advanced symptomatic disease should be treated with systemic therapy with the goal of reducing or reversing symptoms, lymphedema, or threat to organ function. Complete remissions are rare.**
  - **Treatment is typically continued until unacceptable toxicity or plateau in response; maintenance therapy beyond 2 cycles of systemic therapy after determination of plateau is not recommended. If response is then clinically acceptable, patients may be observed (while continuing ART, in PWH). Otherwise, alternative therapy should be initiated.**

[1] Fernandez-Sanchez M, Iglesias MC, Ablanedo-Terrazas Y, et al. Steroids are a risk factor for Kaposi's sarcoma-immune reconstitution inflammatory syndrome and mortality in HIV infection. AIDS 2016;30:909-914.
[2] Guo WX, Antakly T. AIDS-related Kaposi's sarcoma: evidence for direct stimulatory effect of glucocorticoid on cell proliferation. Am J Pathol 1995;146:727-734.
[3] Volkow PF, Cornejo P, Zinser JW, et al. Life-threatening exacerbation of Kaposi's sarcoma after prednisone treatment for immune reconstitution inflammatory syndrome. AIDS 2008;22:663-665.

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PRINCIPLES OF IMMUNE RECONSTITUTION INFLAMMATORY SYNDROME (IRIS)

CLINICAL PRESENTATION/DEFINITION[1]

- **IRIS is characterized by first presentation or paradoxical worsening of pre-existing KS infection following ART initiation.**
- **Typically occurs within 3 months of ART initiation, but possible to occur any time after starting ART.**
- **PWH with advanced KS not on ART at the time of presentation, should start systemic therapy as soon as possible after the initiation of ART to minimize the risk of severe KS-IRIS.**
- **The risk of IRIS is higher in patients with baseline low CD4 T-cell counts and high HIV viral loads.[2]**
- **Clinical manifestations may include development of new lesions or enlargement of existing lesions, worsening lymphadenopathy, or worsening edema. The clinical presentation of IRIS may be challenging to distinguish from the natural history of progressive disease.**
- **If IRIS is suspected, consult with an HIV specialist.**

MANAGEMENT

- **Treatment for KS-IRIS includes systemic chemotherapy and supportive measures; ART should not be discontinued.**
- **Avoid corticosteroids, which may exacerbate KS.**
- **If the patient is on ART only: Symptomatic management; consider addition of systemic chemotherapy (see list of first-line options on KS-F).**
  - **Corticosteroid-sparing treatment such as thalidomide can be considered as an active agent against both KS and IRIS.**
- **If the patient is on ART and chemotherapy: Symptomatic management; consider modification of chemotherapy regimen if progressive.**
- **Use of systemic chemotherapy for extensive disease prior to ART initiation may help prevent KS-IRIS, but this has not been systematically studied.**

PROPOSED CRITERIA FOR DIAGNOSIS OF KS-IRIS[3]

KS-IRIS requires at least one major and one minor criterion:

Major Criteria

- **New onset or enlargement of KS lesion and subsequent regression**
- **Painful lesions**

Minor Criteria

- **Decrease in plasma HIV RNA by >1 $\log^{10}$ copies/mL**
- **Increased blood CD4 T-cell count after ART**

[1] Letang E, Lewis JJ, Bower M, et al. Immune reconstitution inflammatory syndrome associated with Kaposi sarcoma: higher incidence and mortality in Africa than in the UK. AIDS 2013;27:1603-1613.
[2] Panel on Antiretroviral Guidelines for Adults and Adolescents. Guidelines for the Use of Antiretroviral Agents in Adults and Adolescents with HIV. Department of Health and Human Services. Available at https://clinicalinfo.hiv.gov/en/guidelines/adult-and-adolescent-arv. Accessed September 6, 2022.
[3] French MA, Price P, Stone SF. Immune restoration disease after antiretroviral therapy. AIDS 2004;18:1615-1627.

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LOCAL THERAPY

Topical
- **Alitretinoin 0.1% gel[1]**
  - **Apply 2 times daily to affected skin sites; increase to 3–4 times daily if tolerated**
- **Imiquimod, 5% cream[2]**
  - **Apply one sachet to up to 20 cm^2 of affected skin and cover with occlusive dressing for 8 hours three times per week; titrate frequency of application to effect and tolerability (up to once daily), with treatment breaks if tolerability issues occur.**

Radiotherapy[a]
- **Principles of Radiation Therapy (KS-E)**

Cryotherapy
- **Cryotherapy should be done by clinicians with expertise in cutaneous cancer cryotherapy.**

For local control of small, symptomatic lesions, the following may be considered:
- **Marginal excision[b]**
- **Electrodesiccation and curettage**
- **Intralesional chemotherapy**
  - ◊ **Pain from injection is common and may persist for several days. Nonsteroidal anti-inflammatory drugs (NSAIDs) may be useful to relieve pain from injection.**
  - ◊ **Intralesional chemotherapy to plantar and palmar surfaces might be useful in selected cases, but should be approached with caution.**
  - ◊ **Treatment schemas are listed below as a guide. Other treatment schemas have been studied, with a variety of chemotherapy concentrations, doses, administration volumes, frequencies of administration, and total doses/volumes administered. See Discussion for additional references and information.**
  - **Vinblastine[3]**
    - ◊ **0.2 mg/mL solution with a volume of 0.1 mL per 0.5 cm^2 of lesion**
  - **Bleomycin[4]**
    - ◊ **1.5 to 3 units/mL solution with a dose of 0.5 unit per 0.5 cm^2 of lesion (volume given depends on the size of the lesion) every 3 to 4 weeks until response**

Footnotes

[a] For larger lesions when systemic therapy is not feasible or effective, radiotherapy is preferred over excision.
[b] Wide excision is not indicated.

References

[1] Bodsworth NJ, Bloch M, Bower M, et al. Phase III vehicle-controlled, multi-centered study of topical alitretinoin gel 0.1% in cutaneous AIDS-related Kaposi's sarcoma. Am J Clin Dermatol 2001;2:77-87.
[2] Schartz NEC, Chevret S, Paz C, et al. Imiquimod 5% cream for treatment of HIV-negative Kaposi's sarcoma skin lesions: a phase I to II open-label trial in 17 patients. J Am Acad Dermatol 2008;58:585-591.
[3] Epstein JB. Treatment of oral Kaposi sarcoma with intralesional vinblastine. Cancer 1993;71:1722-1725.
[4] Poignonec S, Lachiver LD, Lamas G, et al. Intralesional bleomycin for acquired immunodeficiency syndrome-associated cutaneous Kaposi's sarcoma. Arch Dermatol 1995;131:228.

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PRINCIPLES OF RADIATION THERAPY

- **General Principles**
  - **For most skin lesions, electrons or superficial x-rays can be used to deliver optimal dosimetry and minimize dose to underlying structures. To ensure sufficient dose is delivered for deeper or larger lesions, conformal photon therapy or mixed photon-electron treatment plans may be used. Intensity-modulated RT (IMRT) with or without image guidance may be useful for larger, deeper disease, or disease located anatomically adjacent to critical structures.**
  - **The use of bolus may be necessary to achieve adequate skin dose.**
  - **RT to plantar and palmar surfaces might be useful in selected cases. However, high doses should be approached with caution because of theoretical toxicity concerns like long-term wound healing, particularly in the setting of coexisting lymphedema.**
  - **Risk of secondary cancer, severe or worsening lymphedema, and long-term wound healing complications may be increased after radiation; however, toxicity can be mitigated by utilizing lower-dose radiotherapy regimens.**
    Caution should be exercised with the use of RT to sites of pre-existing lymphedema. In the setting of advanced cutaneous disease, radiation should be reserved for circumstances when systemic therapy is not feasible with the goal of palliation or short-term disease management until systemic therapy may be delivered.

- **General Treatment Information**
  - **Dosing Prescription Regimen[1,2,3]**
    - ◊ **Various dosing schemas may be used. Lower total doses are preferred for smaller and more superficial lesions and palliative situations. Higher doses may be preferred for more extensive, deeply invasive lesions, or when a more durable local response is desired. More fractionated regimens may be preferred for sites with adjacent radiosensitive structures, such as the oral cavity. Examples:**

Fractionated Dose	Biologically Effective Dose [BED]10[a]	BED3
6–8 Gy in 1 fraction[4]	9.6–14.4 Gy	18–29 Gy
20 Gy in 5 fractions[1,3]	28 Gy	47 Gy
24 Gy in 12 fractions[1]	28.8 Gy	40 Gy
30 Gy in 10–15 fractions[2,3]	36–39 Gy	50–60 Gy
40 Gy in 20 fractions[5]	48 Gy	67 Gy

Footnote

[a] BED is a measure of the radiosensitivity of the tissue, with BED10 denoting the estimated effective dose to rapidly growing tumor tissue and BED3 denoting the estimated dose to normal tissue (organs at risk).

References

1 Singh NB, Lakier RH, Donde B. Hypofractionated radiation therapy in the treatment of epidemic Kaposi sarcoma – a prospective randomized trial. Radiother Oncol 2008;88:211-216.
2 Hauerstock D, Gerstein W, Vuong T. Results of radiation therapy for treatment of classic Kaposi sarcoma. J Cutan Med Surg 2009;13:18-21.
3 Kirova YM, Belembaogo E, Frikha H, et al. Radiotherapy in the management of epidemic Kaposi's sarcoma: a retrospective study of 643 cases. Radiother Oncol 1998;46:19-22.
4 Tsao MN, Sinclair E, Assaad D, et al. Radiation therapy for the treatment of skin Kaposi sarcoma. Ann Palliat Med 2016;5:298-302.
5 Stelzer KJ, Griffin TW. A randomized prospective trial of radiation therapy for AIDS-associated Kaposi's sarcoma. Int J Radiat Oncol Biol Phys 1993;27:1057-1061.

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SYSTEMIC THERAPY[a,b,c]

First-line systemic therapy options

For non-transplant KS
- **Preferred**
  - **Liposomal Doxorubicin[d]**
- **Other recommended**
  - **Paclitaxel**

For transplant KS
- **Preferred**
  - **Sirolimus[e]**
- **Other recommended**
  - **Liposomal Doxorubicin[d]**
  - **Paclitaxel**

Relapsed/ refractory systemic therapy[f] options (advanced disease)

If tolerated and durable response (≥3 months) to first-line systemic therapy → **Repeat of first-line systemic therapy or Alternate first-line systemic therapy** → **Progression**

If no response to first-line systemic therapy → **Alternate first-line systemic therapy[g]** → **Progression**

- **Subsequent systemic therapy options for relapsed/ refractory therapy[h] (in alphabetical order)**
  - ◊ **Nivolumab[i]**
  - ◊ **Pembrolizumab[i]**
  - ◊ **Pomalidomide**
- **Useful in certain circumstances (in alphabetical order)**
  - ◊ **Albumin-bound Paclitaxel (if Paclitaxel intolerant)**
  - ◊ **Bortezomib**
  - ◊ **Etoposide**
  - ◊ **Gemcitabine**
  - ◊ **Imatinib**
  - ◊ **Ipilimumab + Nivolumab[i]**
  - ◊ **Lenalidomide**
  - ◊ **Sirolimus[e] (for transplant KS)**
  - ◊ **Thalidomide (for patients with IRIS)**
  - ◊ **Vinorelbine**

Systemic Therapy Dosing (KS-F 2 of 4)

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SYSTEMIC THERAPY DOSING[a,b,c,j]

FIRST-LINE SYSTEMIC THERAPY DOSING

Preferred

- **Liposomal Doxorubicin[d]**
  - **20 mg/m² IV every 2 to 3 weeks**
- **Sirolimus (for transplant KS)**
  - **Loading dose 0.15 mg/kg PO followed by 0.04–0.06 mg/kg/day to maintain trough blood levels of 6–10 ng/mL or 2 mg PO daily (adjust to maintain trough levels of 6–10 ng/mL)**
  - **For PWH on ART, providers should consult with an infectious disease (ID) pharmacist prior to dosing Sirolimus**

Other recommended

- **Paclitaxel**
  - **Dose schedule options:**
    - ◊ **60 mg/m² IV weekly**
    - ◊ **100 mg/m² IV every 2 weeks**
    - ◊ **100 mg/m² IV every 3 weeks**
    - ◊ **135 mg/m² IV every 3 weeks**

SUBSEQUENT SYSTEMIC THERAPY OPTIONS FOR RELAPSED/REFRACTORY THERAPY DOSING

- **Nivolumab[i,k]**
  - **480 mg IV every 4 weeks**
- **Pembrolizumab[i]**
  - **200 mg IV every 3 weeks for up to 6 months**
- **Pomalidomide[l]**
  - **4 or 5 mg/day PO for 21 days of each 28-day cycle[m]**

Useful in certain circumstances (in alphabetical order)

- **Albumin-bound Paclitaxel (if Paclitaxel intolerant)**
  - **100 mg IV on days 1, 8, and 15 of each 28-day cycle**
- **Bortezomib**
  - **1.6 mg/m² IV/SC on days 1, 8, and 15 of each 28-day cycle**
- **Etoposide**
  - **50 mg/day PO for 7 days of each 14-day cycle. After 2 cycles, escalate dose to 100 mg/day PO for 7 days of each 14-day cycle in patients without PR or CR and no toxicity >Grade 2. Dose can be further escalated to 150 mg/day based on tolerance and response**
- **Gemcitabine**
  - **1000 mg/m² IV every 2 weeks**
  - **Or 1000 mg/m² IV on days 1 and 8 every 21 days**
- **Imatinib**
  - **400 mg/day PO**
- **Ipilimumab + Nivolumab[i]**
  - **Nivolumab 240 mg IV every 2 weeks and Ipilimumab 1 mg/kg IV every 6 weeks**
- **Lenalidomide[l]**
  - **25 mg/day PO for 21 days of each 28-day cycle**
- **Sirolimus (for transplant KS)**
  - **Loading dose 0.15 mg/kg PO followed by 0.04–0.06 mg/kg/day to maintain trough blood levels of 6–10 ng/mL or 2 mg PO daily (adjust to maintain trough levels of 6–10 ng/mL).**
  - **For PWH on ART, providers should consult with an ID pharmacist prior to dosing Sirolimus**
- **Thalidomide (for patients with IRIS)[l]**
  - **200 mg/day orally (starting dose, titrated to effect and tolerability)**
- **Vinorelbine**
  - **30 mg/m² IV every 2 weeks**

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SYSTEMIC THERAPY
FOOTNOTES

[a] References for regimens on KS-F (4 of 4).
[b] Corticosteroid pre-medication as an anti-emetic, or to prevent infusion/hypersensitivity reaction (except in cases of prior reaction) should generally be avoided.
[c] An FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN guidelines.
[d] Due to potential risk of cardiotoxicity, perform echocardiogram prior to initial course of Liposomal Doxorubicin and repeat periodically. Consider limiting lifetime dose per prescribing guidelines; however, other data suggest that the patients who need continued treatment may be safely treated beyond 1000 mg/m^2.
[e] For KS associated with immunosuppression from solid organ transplant switching to Sirolimus for immunosuppression may be sufficient for KS control and treatment. Systemic therapy given concurrently with Sirolimus would be appropriate in more aggressive or advanced disease.
[f] Consider repeating any prior systemic therapy that was tolerated and resulted in a durable response.
[g] If both first-line options have already been given, the patient should proceed to subsequent systemic therapy options.
[h] Patients can continue through all treatment options listed, and treatments can be repeated if tolerated and response was durable (≥3 months). In select cases, best supportive care may be an appropriate option.
[i] Immune checkpoint inhibitors (ICIs) should not be used in patients with MCD or KICS due to the risk of flare of these conditions. If the patient has a history of KSHV-associated diseases, ICIs should be used with caution and with consideration of more frequent monitoring for signs and symptoms of KICS or MCD. See NCCN Guidelines for Cancer in People with HIV, Principles of Systemic Therapy and Drug-Drug Interactions. See Discussion.
[j] For PWH, see NCCN Guidelines for Cancer in People with HIV, Principles of Systemic Therapy and Drug-Drug Interactions (HIV-B).
[k] Nivolumab and hyaluronidase-nvhy subcutaneous injection may be substituted for IV Nivolumab. Nivolumab and hyaluronidase-nvhy has different dosing and administration instructions compared to IV Nivolumab.
[l] Patients of childbearing potential treated with Pomalidomide, Lenalidomide, or Thalidomide require effective contraception if participating in sexual activity that could result in pregnancy. Refer to the risk evaluation and mitigation (REMS) program for the specific therapy.
[m] The clinical trial for Pomalidomide used a dose of 5 mg/day. However, Pomalidomide is provided in a 4-mg dose and the NCCN Panel believes that this is a sufficient dose.

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ABBREVIATIONS

ART	antiretroviral therapy
BED	biologically effective dose
CBC	complete blood count
CMP	comprehensive metabolic panel
CR	complete response
DDI	drug-drug interaction
EGD	esophagogastroduodenoscopy
GI	gastrointestinal
HHV-8	human herpesvirus 8
HIV	human immunodeficiency virus
HLF	hemophagocytic lymphohistiocytosis
ICI	immune checkpoint inhibitor
ID	infectious disease
IHC	immunohistochemistry
IL	interleukin
IMRT	intensity-modulated radiation therapy
IRIS	immune reconstitution inflammatory syndrome
KICS	KSHV–associated inflammatory cytokine syndrome
KS	Kaposi sarcoma
KSHV	Kaposi sarcoma-associated herpesvirus
MCD	multicentric Castleman disease
NHL	non-Hodgkin lymphoma
NSAID	nonsteroidal anti-inflammatory drug
OI	opportunistic infection
PD	progressive disease
PEL	primary effusion lymphoma
PR	partial response
PWH	people with HIV
REMS	risk evaluation and mitigation strategy
SD	stable disease
SPEP	serum protein electrophoresis
TAFRO	thrombocytopenia, anasarca, fever, retiniculin fibrosis, and organomegaly
TB	tuberculosis

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NCCN Categories of Evidence and Consensus	
Category 1	Based upon high-level evidence (≥1 randomized phase 3 trials or high-quality, robust meta-analyses), there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2A	Based upon lower-level evidence, there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2B	Based upon lower-level evidence, there is NCCN consensus (≥50%, but <85% support of the Panel) that the intervention is appropriate.
Category 3	Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.

All recommendations are category 2A unless otherwise indicated.

NCCN Categories of Preference	
Preferred	Interventions that are based on superior efficacy, safety, and evidence; and, when appropriate, affordability.
Other recommended	Other interventions that may be somewhat less efficacious, more toxic, or based on less mature data; or significantly less affordable for similar outcomes.
Useful in certain circumstances	Other interventions that may be used for selected patient populations (defined with recommendation).

All recommendations are considered appropriate.

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NCCN Guidelines Version 2.2026
Kaposi Sarcoma

Discussion

This discussion corresponds to the NCCN Guidelines for Kaposi Sarcoma. Last updated: September 16, 2025

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Overview

Kaposi sarcoma is a multifocal malignancy of endothelial cells, which presents with characteristic purpuric, red, or brown macules, papules, and plaques. Four types of Kaposi sarcoma have been described: HIV-associated, classic, iatrogenic or transplant-associated, and endemic.[1-3] Kaposi sarcoma is universally associated with Kaposi sarcoma-associated herpesvirus (KSHV) infection (also known as human herpesvirus-8, or HHV-8).[2] Serologic confirmation of KSHV infection is present in 95% to 98% of patients with Kaposi sarcoma.[2,3] KSHV infections are usually asymptomatic. Like other herpesviruses, KSHV establishes latent infection in humans that is lifelong. Hence, persons with KSHV infection are at lifelong risk of Kaposi sarcoma and other KSHV-associated diseases, including KSHV-inflammatory cytokine syndrome (KICS), KSHV-associated multicentric Castleman disease (MCD) and primary effusion lymphoma (PEL). Immunosuppression of various forms (ie, immunosenescence, human immunodeficiency virus [HIV]-associated and iatrogenic) is recognized to be an important factor in the pathogenesis of Kaposi sarcoma.

The NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Kaposi Sarcoma provide treatment recommendations for patients with HIV-associated, classic, iatrogenic, and endemic Kaposi sarcoma and are intended to assist health care providers with clinical decision-making. This Discussion section provides an overview of the literature supporting the recommendations included in the guidelines. The Panel also publishes separate NCCN Guidelines for Cancer in People with HIV (available at www.NCCN.org), which give recommendations regarding HIV management during cancer therapy, drug-drug interactions (DDIs), radiation therapy, and supportive care for people with HIV (PWH) who have cancer. Recommendations for the management of Kaposi sarcoma for patients in Sub-Saharan Africa can be found in the International Adaption of the NCCN Guidelines for Sub-Saharan Africa for Kaposi Sarcoma (available at https://www.nccn.org/global/what-we-do/international-adaptations).

Guidelines Update Methodology

The complete details of the Development and Update of the NCCN Guidelines are available on the NCCN website at www.NCCN.org.

Literature Search Criteria

Prior to the update of the NCCN Guidelines® for Kaposi Sarcoma, an electronic search of the PubMed database was performed to obtain key literature in Kaposi Sarcoma published since the previous Guidelines update, using the search term: Kaposi Sarcoma. The PubMed database was chosen because it remains the most widely used resource for medical literature and indexes peer-reviewed biomedical literature.[4]

The search results were narrowed by selecting studies in humans published in English. Results were confined to the following article types: Case Reports; Clinica Study; Clinical Trial, Phase II; Clinical Trial, Phase III; Clinical Trial, Phase IV; Controlled Clinical Trial; Evaluation Study; Guideline; Multicenter Study; Observational Study; Practice Guideline; Randomized Controlled Trial; Meta-Analysis; Systematic Reviews; and Validation Studies.

The data from key PubMed articles as well as articles from additional sources deemed as relevant to these guidelines as discussed by the Panel during the Guidelines update have been included in this version of the Discussion section. Recommendations for which high-level evidence is lacking are based on the Panel's review of lower-level evidence and expert opinion.

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Sensitive/Inclusive Language Usage

NCCN Guidelines strive to use language that advances the goals of equity, inclusion, and representation.[5] NCCN Guidelines endeavor to use language that is person-first; not stigmatizing; anti-racist, anti-classist, anti-misogynist, anti-ageist, anti-ableist, and anti-weight-biased; and inclusive of individuals of all sexual orientations and gender identities. NCCN Guidelines incorporate non-gendered language, instead focusing on organ-specific recommendations. This language is both more accurate and more inclusive and can help fully address the needs of individuals of all sexual orientations and gender identities. NCCN Guidelines will continue to use the terms men, women, female, and male when citing statistics, recommendations, or data from organizations or sources that do not use inclusive terms. Most studies do not report how sex and gender data are collected and use these terms interchangeably or inconsistently. If sources do not differentiate gender from sex assigned at birth or organs present, the information is presumed to predominantly represent cisgender individuals. NCCN encourages researchers to collect more specific data in future studies and organizations to use more inclusive and accurate language in their future analyses.

HIV-Associated Kaposi Sarcoma

Historically, when Kaposi sarcoma occurred in the setting of HIV seropositivity, it was considered an AIDS-defining illness and was referred to as AIDS-related or epidemic Kaposi sarcoma. The risk for Kaposi sarcoma in the setting of HIV was previously reported to be increased as much as 3640-fold over the general U.S. population,[6-10] but this risk has declined in the antiretroviral therapy (ART) era.[6,11-14] Still, estimates indicate that the risk of Kaposi sarcoma in PWH between 2000 and 2015 was elevated approximately 498- to 512-fold compared with the general U.S. population,[11,15] and Kaposi sarcoma accounted for approximately 7% of cancers diagnosed in PWH, with an estimated 540 cases diagnosed in the United States in 2020.[16] Five-year survival of patients with HIV-related Kaposi sarcoma has improved in the post-ART era, from 12.1% in 1980 to 1995 to as high as 88% in the post-ART era.[17-19]

When immunosuppression is advanced, HIV-associated Kaposi sarcoma is more common, more aggressive, and more likely to involve viscera, mucosa, and lymph nodes and to cause lymphedema than when immunosuppression is minimal. In fact, CD4+ T-cell counts and HIV viral load correlate with the risk of Kaposi sarcoma development in PWH, and effective ART lowers that risk.[20] Evidence also suggests that ART improves prognosis of Kaposi sarcoma in PWH.

HIV-associated Kaposi sarcoma can also occur in PWH with normal CD4+ T-cell counts and undetectable HIV viral load. Like classic Kaposi sarcoma, these cases commonly occur with advanced age. Even with well-controlled HIV, studies have demonstrated more rapid immunosenescence in PWH than people without HIV, which may permit Kaposi sarcoma to become clinically apparent in KSHV-seropositive PWH. Typically presenting with more limited disease than in PWH with advanced immunosuppression, these cases have prompted a shift toward favoring the term “HIV-associated Kaposi sarcoma” over “AIDS-related Kaposi sarcoma” considering its occurrence in PWH with otherwise preserved immune function as well as in those with advanced immunosuppression.

HIV management during treatment of Kaposi sarcoma in PWH is critical. Co-management by oncology and HIV specialists is recommended for the duration of therapy. Oncology and HIV specialists, along with both an oncology pharmacist and HIV pharmacist if available, should review proposed cancer therapy, supportive care medications, and ART for possible DDIs and overlapping toxicities prior to initiation of therapy for PWH. The NCCN Guidelines for Cancer in People with HIV (available at www.NCCN.org) provide additional recommendations on HIV screening,

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linkage to HIV care, prevention of opportunistic infections, DDIs, radiation therapy, and supportive care for PWH.

Classic Kaposi Sarcoma

Classic Kaposi sarcoma generally involves indolent cutaneous lesions, often of the lower extremities, which may wax and wane or slowly progress over years to decades. It is most common in people of Mediterranean, Eastern European, Middle Eastern, and/or Jewish origins, with a mean age of 74 years at diagnosis.[1,3] It is reported to be 7 to 15 times more common in males than in females.[1,3] Synchronous primary malignancies may be fairly common in patients with classic Kaposi sarcoma, and are more likely to be the cause of death in these patients.[3]

Iatrogenic Kaposi Sarcoma

When Kaposi sarcoma occurs in the context of immunosuppressive therapy (for organ transplant or other reasons), it is called iatrogenic or transplant-associated Kaposi sarcoma.[1] Lesions often appear 2 to 8 months after initiation of immunosuppression, and occur 2 to 3 times more often in males.[1] Although this form of Kaposi sarcoma can be aggressive and involve lymph nodes, mucosa, and/or visceral organs, it frequently responds to cessation of immunosuppression. When this is not feasible, reduction of immunosuppression or change to use of mTOR inhibitors as the immunosuppressive therapy can improve control of Kaposi sarcoma.[21] More information on iatrogenic Kaposi sarcoma is included below (see *Optimizing Immune Function*).

Endemic Kaposi Sarcoma

Endemic Kaposi sarcoma occurs in children and younger adults (<40 years of age) of equatorial Africa and may also occur in immigrants from the region. It is usually more aggressive than classic Kaposi sarcoma, sometimes with visceral, bone, and/or lymph node involvement, although it often begins as skin lesions that remain indolent for several years.[1] As with classic Kaposi sarcoma, endemic disease is 10 to 17 times more common in males than in females.[1]

Other KSHV-associated conditions

Additional KSHV-associated conditions may arise synchronously or asynchronously with Kaposi sarcoma, including PEL, MCD, and KICS.[22-24] As each of these conditions typically follow an aggressive clinical course requiring management beyond that indicated for Kaposi sarcoma alone, it is important that providers caring for persons with Kaposi sarcoma are aware of the presenting features and management considerations for these additional KSHV-associated conditions, as discussed below.

Primary Effusion Lymphoma

PEL is a rare, aggressive CD-20-negative B-cell lymphoma associated with immunosuppression, particularly in the setting of advanced HIV infection. It presents with malignant effusions of clonal CD38+, KSHV+ B-cells, with or without lymphadenopathy. Among PWH, a history of Kaposi sarcoma is associated with 59-fold higher PEL incidence compared with those without a history of any diagnosis historically defined as *AIDS-defining*.[25] Poor performance status, advanced stage disease, and lack of ART treatment of HIV prior to the diagnosis of PEL have been identified as prognostic factors for shorter survival. A PEL-PS model was recently proposed, finding that the combination of ECOG performance status >2 and hemoglobin <8 g/dL was associated with poor survival rates in HIV-associated PEL.[26] Cytology and flow cytometry of effusion fluid can be used to evaluate for coexisting PEL in patients with Kaposi sarcoma. More information on PEL can be found in the NCCN Guidelines for B-Cell Lymphomas, under HIV-Related B-Cell Lymphomas (available at www.NCCN.org).

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KSVH-Associated MCD and KICS

Unexplained fevers occurring in the context of Kaposi sarcoma, particularly when associated with lymphadenopathy/splenomegaly and cytopenias, should prompt workup of MCD and KICS, with imaging, C-reactive protein, KSHV serum viral load, serum protein electrophoresis (SPEP), IL-6, and IL-10.[22-24]

KSHV-Associated MCD

MCD is classified as tumor-like lesion with B-cell predominance, which may be idiopathic or associated with KSHV.[27] KSHV-associated MCD is a remitting-relapsing disease that typically follows an aggressive course. Immunocompromising conditions are a primary risk factor for KSHV-associated MCD, with HIV being the most common.[28] Clinical presentation includes multifocal lymphadenopathy, splenomegaly, signs of fluid accumulation (edema/effusions), cytopenias, liver and kidney dysfunction, and constitutional symptoms (fevers, night sweats) driven by cytokines such as IL-6.[29-31] KSHV viremia is typically marked during flares of MCD (≥4 log) and undetectable or low level during remissions (~3 log).[31] More information on MCD is available in the NCCN Guidelines for Castleman Disease (available at www.NCCN.org).

KICS

KICS and KSHV-associated MCD share clinical features of an inflammatory syndrome with often dramatic KSHV viremia.[22-24] KICS is a diagnosis of exclusion, distinguished from MCD and lymphoma by absence of pathologic characteristics of either upon biopsy of lymphadenopathy or other masses.[22] KICS may also mimic Kaposi sarcoma-associated immune reconstitution inflammatory syndrome (KS-IRIS), but is distinguished from it by not being temporally tied with initiation of ART as expected with IRIS.

The working definition of KICS proposed by Polizotto et al involves the presence of specific clinical symptoms, evidence of inflammation and KSHV viremia, and the exclusion of MCD when lymphadenopathy is present (see Table 1).[22] Lymphadenopathy in KICS is often less marked than can be seen with MCD, and there are no specific histologic features currently associated with KICS.

Optimal management of KICS is under investigation, with current standards paralleling that of KSHV-associated MCD. Liposomal Doxorubicin as Kaposi sarcoma-directed therapy with Rituximab has resulted in control of KICS.[32] This regimen is therefore the preferred option for patients with KICS. Other Kaposi sarcoma-directed therapy with Rituximab may also be considered. In mild cases, Kaposi sarcoma-directed therapy alone may be sufficient, and Rituximab may be added later if disease response is inadequate. For transplant recipients, the Sirolimus component of immunosuppression should be maximized and other components should be minimized. Early consultation with Kaposi sarcoma experts is encouraged.

If inflammatory syndrome and lymphadenopathy/organomegaly improve, Rituximab can be withheld while Kaposi sarcoma-directed therapy is continued for 3 months or for as long as needed for adequate/optimal Kaposi sarcoma response. Rituximab should be resumed if inflammatory symptoms or lab markers flare after Rituximab is stopped.

For refractory cases, management strategies like that used for KSHV-MCD may be effective. Thus, the addition of anti-IL-6 directed therapy (ie, Tocilizumab or Siltuximab) to Kaposi sarcoma-directed therapy should be considered.[33-35]

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Management of Kaposi Sarcoma

Diagnosis and Workup of Kaposi Sarcoma

Multiple clinical and histologic presentations of Kaposi sarcoma have been described. Mucosal and cutaneous lesions may be characterized clinically as papules, plaques, nodules (sometime pedunculated), and bullae. Large plaques may form from coalescence of smaller plaques or nodules and may ulcerate or develop bullae. Hyperpigmented macules (lacking palpable change in skin thickness) rarely represent active disease; rather, they are very common after lesion regression due to residual hemosiderin-related pigmentation. Histologic subtypes include anaplastic, telangiectatic, lymphedematous, hyperkeratotic, keloidal, micronodular, pyogenic granuloma-like, ecchymotic, and intravascular variants of Kaposi sarcoma.

Lymphedema is a common complication of Kaposi sarcoma and may also be a predisposing factor to the development of Kaposi sarcoma. Lymphedema can be caused by not only nodal involvement but involvement of lymphatic vessels. Hyperkeratotic variants with verrucous and hyperkeratotic changes are notably associated with chronic and severe Kaposi sarcoma-associated lymphedema and may require deeper biopsy to confirm the presence of Kaposi sarcoma.

As described in the guidelines above, Kaposi sarcoma is diagnosed by pathology and immunophenotyping. Workup should include evaluation for history of immunosuppressive condition or therapy, such as transplant, local or systemic glucocorticoids (including topical, intranasal, inhaled, intraarticular, as well as systemic), or other immunosuppressive therapies. Complete skin, oral, and lymph node exams, including evaluation of edema and photography of oral, conjunctival, and cutaneous lesions for evaluation and monitoring of extent of disease are recommended. It is important to note that certain opportunistic infections can result in cutaneous lesions that can mimic Kaposi sarcoma lesions (eg, bacillary angiomatosis, blastomycosis, cryptococcosis).[36-39] Therefore, in addition to biopsy of suspected lesions, involvement of an HIV specialist to evaluate for coexisting opportunistic infections is appropriate in PWH and those with advanced immunosuppression. Laboratory evaluation should include complete blood count (CBC) with differential, comprehensive metabolic panel, HIV screening, and/or diagnostic testing.

For PWH, care coordination between the HIV specialist and the oncology team is recommended. Recommendations for the management of cancer in PWH, including the possible need to modify ART due to DDIs, are presented in the NCCN Guidelines for Cancer in People with HIV (available at www.NCCN.org). All PWH should have recent T-cell subsets including quantitative CD4+ T-cell counts and HIV viral load to assess immune function and HIV control.

Depending on symptoms and findings that may be concerning for visceral or bone involvement, additional workup may be necessary. This may include fecal occult blood testing (FOBT), chest x-ray, chest/abdomen/pelvis CTs with contrast, MRI with and without contrast, bronchoscopy, upper endoscopy, or colonoscopy.

In the event of suspected coexisting KSHV-associated PEL, MCD, or KICS, additional workup is necessary, including imaging, such as fluorodeoxyglucose (FDG)-PET/CT scans, the laboratory workup noted above, cytology with flow cytometry of effusions, and biopsy of lymphadenopathy (excisional biopsy is optimal).

In persons with HIV-associated Kaposi sarcoma, it is important to note that interpretation of imaging in PWH who have cancer is complicated by the increased incidence of non-malignant lesions that may be mistaken for cancer spread or recurrence. Opportunistic infections in the lung include *Mycobacterium tuberculosis* (TB), cytomegalovirus (CMV), invasive fungal infections, and *Pneumocystis jirovecii* pneumonia (PCP).[40] Furthermore,

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non-infectious, non-malignant pulmonary manifestations of HIV can be difficult to interpret on imaging studies, including interstitial pneumonia and granulomatous disease.[40,41] Additionally, brain lesions seen in PWH may result from opportunistic infections, such as viral encephalitis, aspergillosis, toxoplasmosis, cryptococcosis, bacterial meningitis, tuberculosis, progressive multifocal leukoencephalopathy, and mycobacterium avium complex (MAC).[42,43] PWH are at increased risk of lymphomas including primary central nervous system lymphoma. Benign non-infectious brain lesions can also occur in PWH (eg, vascular complications, hydrocephalus).[42,43] Similarly, immune response to HIV and opportunistic infections commonly cause lymphadenopathy in PWH, which can be seen on CT and may be avid on F-18 FDG PET/CT.[44,45] Non-malignant causes of lymphadenopathy are more marked in patients with higher HIV viral loads and lower CD4+ T-cell counts.[46] Therefore, involvement of an HIV specialist to assist with determining the need for a workup of suspected concurrent opportunistic infection or HIV-associated complications may be appropriate.

Staging of Kaposi Sarcoma

As delineated in the guidelines above, Kaposi sarcoma is staged using a TIS system in which aspects of the tumor (T), immune system (I), and systemic disease (S) are assessed with a 0 for good risk and 1 for poor risk.[47] Data have shown that the I stage has less prognostic value than the T or S stages for PWH on ART.[18] Kaposi sarcoma staged as T1S1 appears to have the worst prognosis. In a study of 211 patients with HIV-related Kaposi sarcoma, those staged as T1S1 had a 3-year survival rate of 53%, whereas for those staged as T0S0, T1S0, or T0S1, the 3-year survival rates were 88%, 80%, and 81%, respectively (P = .0001).[18] Notably, Kaposi sarcoma management, as outlined in these guidelines, is primarily stratified by limited cutaneous versus advanced disease (ie, T1 disease, extensive T0 cutaneous disease, or nodal disease), reflecting the tumor (T) component of the TIS staging system.

Assessing Response of Kaposi Sarcoma

Response of Kaposi sarcoma to therapy has been formally defined by the AIDS Clinical Trials Group (ACTG) Oncology Committee as follows[47]:

- Complete response (CR) is defined as the absence of any detectable residual disease, including tumor-associated (local) edema, persisting for at least 4 weeks. Patients known to have had visceral disease should have restaging with appropriate endoscopic or radiographic procedures relevant to sites involved at baseline.
- Partial response (PR) is defined as no new mucocutaneous lesions, visceral sites of involvement, or the appearance or worsening of tumor-associated edema or effusions; AND
  - A ≥50% decrease in the number of all previously existing lesions lasting for at least 4 weeks; OR
  - Complete flattening of at least 50% of all previously raised lesions (ie, 50% of all previously nodular or plaque-like lesion become macules); OR
  - A 50% decrease in the sum of the products of the largest perpendicular diameters of at least five measurable lesions.
  - Note that when there is residual tumor-associated edema or effusion, but disease otherwise meets criteria for CR, response should be classified as partial.
- Stable disease (SD) is defined as any response that does not meet the criteria for progressive disease (PD) or PR.
- PD is defined as an increase of ≥25% in the size of pre-existing lesions and/or the appearance of new lesions or sites of disease and/or a change in the character of the skin or oral lesions from macular to plaque-like or nodular of ≥25%. If new or increasing tumor-associated edema or effusion develop, disease is considered progressive.

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Many Kaposi sarcoma lesions that are responsive to therapy will flatten and change color but remain pigmented (ranging from very dark brown to tan) as non-palpable macular skin lesions. Biopsy of these will often identify residual siderophages and/or free hemosiderin pigment in the tissue without tumor cells. This is attributed to long-term iron deposition resulting from red blood cell extravasation into the tissue (dermal layer characteristically in cutaneous lesions). Care should be taken to distinguish this “tattoo” effect from active disease, as additional therapy is not indicated for the former. Over time, many lesions will gradually fade, depending on many factors, including chronicity and size of the lesion as well as degree to which red blood cell extravasation occurred.

Optimizing Immune Function

For all types and severities of Kaposi sarcoma, optimizing immune function is a critical component of management, and in some cases may be the only intervention needed for initial control. In PWH, maximizing HIV suppression and immune reconstitution with appropriate ART is therefore critical.

Iatrogenic immunosuppression—including over-the-counter topical or intranasal, as well as prescription inhaled, locally injected or systemic glucocorticoids—is often identified on a careful medication history in persons with and without HIV prior to initial presentation or flare of Kaposi sarcoma. Rituximab is also associated with initial presentation or flare of Kaposi sarcoma in individuals without HIV, related to the associated activation of KSHV.[48-51] Patients and their providers should be educated on the importance of cessation and avoidance of immunosuppressive therapies whenever possible, including topical, local, and systemic therapies. Otherwise, immunosuppressive therapy should be minimized as much as possible to reduce risk of Kaposi sarcoma progression and flares.[52-54]

Sirolimus is an mTOR inhibitor that functions as an immunosuppressant. mTOR inhibition suppresses Kaposi sarcoma tumors, so–unlike other immunosuppressants–Sirolimus can be an effective treatment against transplant-associated Kaposi sarcoma.[21] Furthermore, when Sirolimus is administered in patients with transplant Kaposi sarcoma, glucocorticoids and other immunosuppressive agents can often be halted or minimized. Sirolimus is discussed in more detail below (see *First-Line Systemic Therapy*).

Limited Cutaneous Kaposi Sarcoma

PWH with limited cutaneous disease that is asymptomatic and cosmetically acceptable to the patient may be treated with ART alone (see below). Persons without HIV can be observed until the disease becomes symptomatic (eg, edema, oral Kaposi Sarcoma, or other manifestation that interfere with normal functions or activities) and/or cosmetically bothersome or progressive. Rapid progression during observation may be an indication for initiation of systemic Kaposi Sarcoma therapy.

PWH with symptomatic and/or cosmetically bothersome limited cutaneous disease should be treated with ART and with the most minimally invasive and least toxic therapy possible. When necessary, a limited number of cycles of systemic therapy (eg, 3–6; options discussed below) may be sufficient for those initiating or re-initiating ART. Other first-line therapy options include enrollment in a clinical trial or local therapy.

Local therapies include topical treatment, radiation therapy, cryotherapy, and for small, symptomatic lesions, marginal excision, electrodessication and curettage, and intralesional chemotherapy may be considered and are discussed in more detail below. Wide excisions with the goal of achieving negative margins are not indicated, as they do not contribute to control of Kaposi sarcoma. For larger lesions when systemic therapy is not feasible or effective, radiotherapy is preferred over excision. Intralesional

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chemotherapy and radiation to plantar and palmar surfaces may be useful in selected cases but should be approached with caution because of toxicity or functional adverse events. Increased risk of long-term wound healing, particularly in the setting of coexisting lymphedema, is a concern.

Initiation of ART may result in IRIS within 3 to 6 months and patients should therefore be reassessed within 4 weeks for signs and symptoms (see *Immune Reconstitution Inflammatory Syndrome*, below).

The same treatment options, without ART, are recommended for patients without HIV who have limited cutaneous disease that is symptomatic and/or cosmetically bothersome.

If progressive, relapsed, or refractory disease is suspected, consider biopsy of lesions since they may be areas of post-inflammatory pigment and/or mimickers of Kaposi sarcoma. If it is Kaposi sarcoma, the individual should be evaluated for inadequate HIV control or other immunosuppressant use, which could be contributing factors to inadequate Kaposi sarcoma control. Change in ART in conjunction with an HIV specialist may be warranted for some cases (see the NCCN Guidelines for Cancer in People with HIV, available at www.NCCN.org).

For PWH with optimized HIV control and those without HIV, Kaposi sarcoma-therapy should be determined by the extent of disease. If Kaposi sarcoma is stable or a response plateau occurs with initial therapy, the patient should be observed (while ART is continued in PWH). If the disease progresses or relapses after an initial response to therapy, repeat use of the previously effective therapy may be considered, particularly if response was durable and the therapy was well tolerated.

Advanced Cutaneous, Oral, Visceral, or Nodal Kaposi Sarcoma

First-line treatment options for patients with advanced cutaneous, oral, visceral, or nodal Kaposi sarcoma are systemic therapy or clinical trial participation, in conjunction with ART for PWH. Patients not immediately eligible for systemic therapy should undergo medical optimization to address reversible issues (eg, active infection requiring antibiotic or antiviral treatment, high-level HIV viremia requiring initiation/modification of ART) so that systemic therapy can be delivered as soon as possible. While radiation therapy may be considered, systemic therapy remains preferred for first-line therapy and relapsed/refractory therapy for disseminated disease whenever feasible. In conjunction, ART should be continued or initiated for PWH. The data supporting these treatment options are described below. Well-designed clinical trials may also be considered for frontline therapy, with the consideration that established systemic therapies are often most appropriate for frontline therapy of *symptomatic* visceral or extensive cutaneous disease (ie, with lymphedema).

Antiretroviral Therapy for People with HIV

Reconstitution of immune function, maintenance of viral suppression, and avoidance of additional immunosuppression are critical to prevent additional Kaposi sarcoma lesions and maintain response to therapy in PWH. In fact, in the setting of limited cutaneous disease, remissions or SD may occur with optimization of immune function and HIV viral suppression alone.[55-61] Therefore, co-management with an HIV specialist to optimize suppression of HIV and reconstitution of immune function with ART is important for patients with HIV-related Kaposi sarcoma (see the NCCN Guidelines for People with HIV, available at www.NCCN.org).

Immune Reconstitution Inflammatory Syndrome

Initiation of ART may result in IRIS within 3 to 6 months in a reported 6% to 39% of patients with HIV-related Kaposi sarcoma.[52,62-65] KS-IRIS is characterized by first presentation or paradoxical worsening of pre-existing Kaposi sarcoma infection following ART initiation.[66] Clinical manifestations of IRIS may be challenging to distinguish from natural history of

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progressive disease and may include the often rapid development of new lesions or enlargement of existing lesions, worsening lymphadenopathy, increased tenderness, and lymphedema. Individuals with pulmonary involvement, concurrent or recent use of glucocorticoids, and/or advanced immunosuppression may be at increased risk.[52,62,64] The risk of IRIS is higher in patients with low baseline CD4 T-cell counts and high HIV viral loads.[67]

PWH with advanced Kaposi sarcoma not on ART at the time of presentation should start systemic therapy as soon as possible after the initiation of ART to minimize the risk of severe KS-IRIS. A randomized open-label clinical trial compared two treatment strategies in patients with mild-to-moderate HIV-related Kaposi sarcoma in sub-Saharan Africa: ART with concurrent low-dose oral etoposide versus ART with etoposide later, as needed.[68] Results demonstrated that early initiation of chemotherapy and ART together decreased early progression of Kaposi sarcoma and IRIS. Early progression of Kaposi sarcoma and IRIS incidences with immediate etoposide versus ART alone were 16% versus 39%, and 7% versus 21%, respectively. Another clinical trial compared immediate versus as-needed oral etoposide in PWH with mild-to-moderate Kaposi sarcoma initiating ART.[69] Immediate oral etoposide delayed Kaposi sarcoma progression (P = .021), decreased suspected KS-IRIS events (P = .003), and decreased time to initial Kaposi sarcoma response (P = .003). However, no long-term clinical benefit was observed compared to the as-needed approach.

KS-IRIS is an indication for urgent initiation of systemic Kaposi sarcoma therapy and should involve coordination with an HIV specialist. Signs of past IRIS may be an indication for earlier initiation of systemic Kaposi sarcoma therapy. ART should not be delayed or discontinued unless life-threatening IRIS develops. In contrast with management of IRIS for some opportunistic infections, glucocorticoids are generally contraindicated in Kaposi sarcoma, as well as in Kaposi sarcoma-associated IRIS, because of the potential for life-threatening Kaposi sarcoma exacerbation resulting from stimulatory effects of glucocorticoids on Kaposi sarcoma spindle cells and association of glucocorticoid use with increased mortality.[52-54]

Corticosteroid-sparing treatment such as immunomodulatory drugs (IMiDs) can be active for both Kaposi sarcoma and IRIS. Whereas there are no prospective trials using thalidomide for Kaposi sarcoma-associated IRIS, successful control of steroid-refractory IRIS with thalidomide has been reported, and the immunomodulatory drugs, including thalidomide, lenalidomide, and pomalidomide, are an active class of therapy in Kaposi sarcoma as discussed further below.[70-73]

Goals of Therapy

It is important to note that individual Kaposi sarcoma lesions may be distinct clones that arise because of the common risk factors of immunosuppression and latent KSHV infection as opposed to metastases. Furthermore, persistence of KSHV infection results in ongoing risk of recurrence/disease progression. Currently, eradication of KSHV is not possible. Therefore, treatment of existing disease may not prevent occurrence of future lesions. Given this and the fact that many presentations of Kaposi sarcoma are not life-threatening, the goals of therapy are focused on disease control. Wide excisions with the goal of achieving negative margins are not indicated, as they do not contribute to control of Kaposi sarcoma. Local therapy (topical, intralesional, cryotherapy, or radiation) or marginal excisions should instead be used for limited symptomatic lesions. For larger lesions, radiotherapy is preferred over excision when systemic therapy is not feasible or effective. Systemic therapy with risk of long-term toxicities should be reserved for more serious presentations. In PWH with limited cutaneous disease that is asymptomatic and cosmetically acceptable, ART alone may allow

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remission or stable disease by optimization of immune function and HIV viral suppression. Patients with advanced disease should be treated with systemic therapy with the goal of reducing or reversing symptoms, lymphedema, or threat to organ function. Complete remissions are rare, particularly in long-standing and or advanced disease.

Local Therapy

Topical Therapies

Topical therapies are an option for patients with limited cutaneous disease that is symptomatic and/or cosmetically bothersome. Alitretinoin gel, a retinoid, was studied in a phase III vehicle-controlled, double-blind, multi-centered study, in which 134 patients with HIV-related Kaposi sarcoma received either 0.1% alitretinoin gel or vehicle gel twice daily for 12 weeks.[74] The cutaneous tumor response rates were 37% in the Alitretinoin group compared with 7% in the control group. Another very similar randomized, multicenter, double-blind, vehicle-controlled study also compared tumor response rates in patients with HIV-related Kaposi sarcoma between an Alitretinoin group and a control group.[75] Response rates in the 268 patients were 35% for those who received 0.1% Alitretinoin gel compared with 18% for those who received the vehicle gel. In both studies, Alitretinoin gel was well tolerated, with mostly mild to moderate adverse events that were limited to the application site and that were relieved when treatment was stopped. Data on the use of Alitretinoin in classic Kaposi sarcoma is limited to two case reports, in which one patient showed significant improvement and the other did not.[76,77]

Imiquimod is a topical immune response modulator with antiviral and antitumor activity.[78] It is used in a variety of skin conditions including malignancies and warts.[78,79] Case reports have shown that Imiquimod cream can be safe and effective in some patients with classic or transplant-associated Kaposi sarcoma.[80-87] In a single-center, open-label, phase I/II trial, 17 patients without HIV who had Kaposi sarcoma received Imiquimod 5% cream 3 times per week for 24 weeks.[88] The response rate was 47%. Over half of the patients reported local itching and erythema, but treatment was generally well tolerated. Imiquimod is not well studied as a treatment for patients with cutaneous HIV-related Kaposi sarcoma.[89,90] The Panel includes Imiquimod as an option for patients with limited cutaneous HIV-related Kaposi sarcoma based on extrapolation from the data presented above in other settings, expert opinion, and non-published anecdotal data.

Radiation Therapy

Kaposi sarcoma is radioresponsive, with CR rates of treated lesions reported in the range of 60% to 93%.[86,91-95] Radiation therapy for Kaposi sarcoma is used in patients with limited cutaneous disease that is symptomatic and/or cosmetically bothersome. For patients with advanced disease, systemic therapy is preferred over radiation therapy in first-line and for relapsed/refractory disease if systemic therapy is feasible based on performance status and medical comorbidities. In the setting of advanced disease, radiation therapy should be reserved for cases where systemic therapy is not feasible, with the goal of palliation or short-term disease management until systemic therapy may be delivered.[96] Radiation therapy may also be used for disease refractory to multiple types of systemic therapy.

When radiation is used, hypofractionated regimens (eg, 20 Gy in 5 fractions) appear to be equally effective as the standard regimen of 24 Gy in 12 fractions.[97,98] Dose fractionation should be based on the site of treatment with consideration for surrounding normal tissue tolerance. Lower total doses are preferred for smaller and more superficial lesions and palliative situations. Higher doses may be preferred for more extensive, deeply invasive lesions, or when a more durable local response is desired. More fractionated regimens may be preferred for sites with adjacent radiosensitive structures, such as the oral cavity.

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The side effects of radiation for Kaposi sarcoma are site-dependent, but typically manageable given the low doses needed to achieve a response.[91-94] Radiation to plantar and palmar surfaces might be useful in selected cases but should be approached with caution because of toxicity or functional adverse events. Still, the risk of secondary cancer, severe or worsening lymphedema, and long-term wound healing complications may be increased after radiation. Early recognition and treatment of dermatitis, oral mucositis, and lymphedema are especially important.[91,93,99] The risk of lymphedema is already elevated in patients with Kaposi sarcoma and may increase after radiation.[100] Therefore, caution should be exercised with the use of radiation to sites of pre-existing lymphedema. Early referral to and co-management with a lymphedema specialist is recommended.

Intralesional Chemotherapy

Intralesional chemotherapy with Vinblastine or Bleomycin is an option for local control of small lesions in patients with limited cutaneous disease that is symptomatic and/or cosmetically bothersome.

Intralesional injection of Vinblastine has been studied in case reports, case series, and one small randomized trial of patients with oral HIV-related Kaposi sarcoma.[101-107] In a large series of 50 PWH with 144 oral Kaposi sarcoma lesions, CR was seen in 74% of lesions and PR in 26%.[104] The recurrence rate was 26%, with a mean disease-free period of 12.9 weeks. Consistent with the safety profile seen in other studies, pain was reported by 72% of participants, ulceration occurred in 22%, and temporary numbness was seen in 12%. Pain is generally mild to moderate and relieved with pain medication, and ulceration is generally self-limiting.

Studies on the use of intralesional Vinblastine injection for *cutaneous* lesions are more limited.[108,109] In a trial of 11 patients with HIV-related Kaposi sarcoma, 88% of cutaneous lesions showed a complete or partial clinical response.[108] Treatment resulted in inflammation and blistering of the lesion prior to healing, and the final results were not cosmetically optimal because of post-inflammation hyperpigmentation. Most patients reported aching pain 6 to 48 hours post-treatment that was relieved with pain medication.

Intralesional Vinblastine has also been used in cutaneous lesions in patients with classic Kaposi sarcoma.[110]

Intralesional Bleomycin has been studied in a limited number of small trials.[111-113] One clinical trial treated 134 lesions in 15 patients with HIV-associated cutaneous Kaposi sarcoma.[113] CR was observed in 58 lesions (43%), PR in 60 lesions (44%), and failure in 6 lesions (4.4%). Recurrence at the site of injection occurred in 17% of partial responders, none after a CR. The only reported side effect was local pain 48 hours after injection in one patient.

Marginal Excision

Marginal excision is an option for local control of small lesions in patients with limited cutaneous disease that is symptomatic and/or cosmetically bothersome. Data regarding outcomes of the excision of cutaneous Kaposi sarcoma lesions are limited and appear to be restricted to individuals without HIV.[114-118] Wide excisions with goal of negative margins is not indicated as this does not contribute to prevention of additional Kaposi sarcoma lesion development as discussed above.

Cryotherapy

Cryotherapy is also an option for patients with limited cutaneous disease that is symptomatic and/or cosmetically bothersome. In a small study of 30 patients, 125 lesions were treated with cryotherapy.[120] Nineteen (63%) patients experienced a CR without recurrence. Blistering was frequent, local pain was limited, and the treatment was well tolerated overall. Other studies also confirm that cryotherapy can be effective in patients with classic Kaposi sarcoma.[87] It is most likely to be effective on non-nodular,

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smaller lesions (ie, <1 cm). Cryotherapy should be done by clinicians with expertise in cutaneous cancer cryotherapy.

Electrodesiccation and Curettage

The Panel also includes electrodesiccation and curettage as an option for local control of small, symptomatic lesions. Data for the use of this technique in patients with Kaposi sarcoma are extremely limited.[119]

Systemic Therapy

Systemic therapy may be appropriate for cases of limited cutaneous Kaposi sarcoma and is the main recommended approach for patients with advanced disease. Specific options are discussed below. Corticosteroid pre-medication as an anti-emetic or to prevent infusion/hypersensitivity reaction (except in cases of prior reaction) should generally be avoided.

First-Line Systemic Therapy

The preferred first-line systemic therapy is Liposomal Doxorubicin for non-transplant Kaposi sarcoma. Paclitaxel is another recommended option. Sirolimus alone may be sufficient for transplant-associated KS, particularly if able to halt or reduce other immunosuppressive agents. It is appropriate to combine sirolimus with other KS directed therapies in severe or refractory cases of transplant-associated KS.

The preference for Liposomal Doxorubicin is supported by several randomized controlled trials. In a randomized phase III trial, 258 patients with advanced HIV-related Kaposi sarcoma were randomized to receive Pegylated-liposomal Doxorubicin or Doxorubicin, Bleomycin, and Vincristine (ABV).[121] The overall response rate was 46% (95% CI, 37%–54%) in the Liposomal Doxorubicin arm and 25% (95% CI, 17%–32%) in the ABV arm. The median time to treatment failure (TTF) was approximately 4 months in both groups. Most patients in both arms experienced ≥1 grade 3–4 adverse event, with leukopenia, nausea/vomiting, anemia, and peripheral neuropathy as the most common adverse events in the Liposomal Doxorubicin group. Pegylated-liposomal Doxorubicin was compared with Bleomycin and Vincristine (BV) in another randomized trial of patients with HIV-related Kaposi sarcoma (n = 241).[122] As in the other trial, response rates were superior in the Liposomal Doxorubicin group compared with the BV group (59% vs. 23%; $P < .001$). Pegylated-liposomal Doxorubicin resulted in an increased risk of neutropenia but was less likely to result in early treatment cessation. Liposomal Doxorubicin has also been shown to have activity in classic and transplant-associated Kaposi sarcoma.[86,123] Another randomized, double-blind trial compared Pegylated-liposomal Doxorubicin to Liposomal Daunorubicin in 60 patients with HIV-related Kaposi sarcoma.[124] Tumor responses were seen in more participants in the Pegylated-liposomal Doxorubicin group than in the group receiving Liposomal Daunorubicin (55.0% vs. 31.6%). Rates of adverse events associated with Pegylated-liposomal Doxorubicin were 30% for neutropenia, 28.3% for nausea, and 16.7% for asthenia.

Liposomal Doxorubicin is associated with risk of cardiotoxicity.[125-127] Therefore, an echocardiogram should be performed prior to initial course of Liposomal Doxorubicin and should be repeated as clinically indicated.[128] Consider limiting lifetime dose per prescribing guidelines; however, some data suggest that the patients who need continued treatment may be safely treated beyond 1000 mg/m^2.[129] Regular echocardiographic monitoring is recommended in such situations.

For Kaposi sarcoma associated with immunosuppression from solid organ transplant, switching to Sirolimus for immunosuppression may be sufficient for Kaposi sarcoma control and treatment. In a study of 15 patients who underwent kidney transplants and developed Kaposi sarcoma, Cyclosporine therapy was stopped and Sirolimus was initiated.[21] At 3 months after the start of sirolimus therapy, all cutaneous lesions had disappeared in all patients, all with confirmed histologic remission at 6

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months. No acute episodes of rejection or changes in kidney-graft function were seen. Minimizing other components of immunosuppression, especially glucocorticoids, is recommended when possible. Adjunctive local Kaposi sarcoma-directed therapy may be added to facilitate or expedite Kaposi sarcoma response. The addition of systemic Kaposi sarcoma-directed therapy is appropriate for more rapidly progressive, advanced cutaneous or visceral Kaposi sarcoma.

An alternative option for first-line systemic therapy for patients with Kaposi sarcoma is Paclitaxel. Early studies showed that it has significant activity in the advanced disease setting, with neutropenia as the most frequent dose-limiting toxicity.[86,130-132] One trial randomized 73 patients with advanced HIV-related Kaposi sarcoma to Paclitaxel or Pegylated-liposomal Doxorubicin.[133] The two arms were statistically equivalent with regard to response rates (56% for Paclitaxel vs. 46% for Pegylated-liposomal Doxorubicin; *P* = .49), median progression-free survival (17.5 vs. 12.2 months; *P* = .66), and 2-year survival (79% vs. 78%; *P* = .75). A trend toward increases in grade 3–5 toxicity was seen in the paclitaxel arm (84% vs. 66%; *P* = .077), with 1 lethal, grade 5 pulmonary embolism in a patient treated with Paclitaxel. A systematic review of randomized trials and observational studies in patients with advanced HIV-related Kaposi sarcoma found no evident differences between Liposomal Doxorubicin, Liposomal Daunorubicin, and Paclitaxel, although the number of studies identified was low.[134] Data on the use of Paclitaxel in non–HIV-related Kaposi sarcoma are more limited.[135,136]

Systemic Therapy of Relapsed/Refractory Disease

At first progression, the same systemic therapy options as in first line (Liposomal Doxorubicin, Sirolimus for transplant Kaposi Sarcoma, or Paclitaxel, discussed above) may be considered as follows:

- If first-line therapy was tolerated and a durable response (>3 months) was seen, then a repeat of the therapy used in first line should be considered.
- If there was no response to first-line systemic therapy, then an alternative first-line therapy option should be given.

Following subsequent progressions, Liposomal Doxorubicin or Paclitaxel, whichever has not yet been administered, is recommended.[137,138] Following treatment with Liposomal Doxorubicin, Sirolimus for transplant Kaposi Sarcoma, and Paclitaxel, the Panel recommends Nivolumab, Pembrolizumab, or Pomalidomide. Other therapies that may be useful for patients with relapsed/refractory Kaposi sarcoma in certain circumstances include Albumin-bound Paclitaxel (if Paclitaxel intolerant), Bortezomib, Etoposide, Gemcitabine, Imatinib, Ipilimumab + Nivolumab, Lenalidomide, Sirolimus (for transplant Kaposi sarcoma; discussed in *First-Line Systemic Therapy*, above), Thalidomide (for patients with IRIS), and Vinorelbine. Patients can continue through all treatment options listed, and treatments can be repeated if they were tolerated and the response was durable (≥3 months). In select cases, best supportive care may be an appropriate option.

Evidence suggests that immune checkpoint inhibitors (ICIs) are generally safe and effective in PWH who have a variety of cancers, including Kaposi sarcoma.[139,140] Case reports and retrospective reviews have shown that Nivolumab monotherapy can be effective in the treatment of refractory Kaposi sarcoma.[141,142] Results of the AIDS Malignancy Consortium 095 Study show that Nivolumab has a similar safety profile in PWH compared with individuals without HIV.[143] In addition, objective partial responses were seen in 6 of the 15 participants who had HIV-related Kaposi sarcoma (40%; 95% CI, 16.3–64.7), regardless of PD-L1 expression. The median duration of response was 12.5 months in the Kaposi sarcoma group.

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A multicenter, phase II, single-arm trial included 17 patients (8 with classic Kaposi sarcoma and 9 with endemic Kaposi sarcoma) treated with Pembrolizumab.[144] The primary endpoint was the best overall response rate within a 6-month timeframe, and three or more responses among the 17 patients were needed for the primary endpoint to be met. Two patients (22%) had a CR, 10 patients had a PR (59%), and 5 patients (29%) had SD, with a best overall response rate of 71% (95% CI, 44–90), meeting the primary outcome. Adverse events occurred in 13 patients, including two grade 3 adverse events (1 acute cardiac decompensation and 1 granulomatous reaction). Another trial, which was a multicenter phase I trial, assessed Pembrolizumab in 32 cisgender men with HIV-associated Kaposi sarcoma on ART.[145] The overall response rate in the 29 evaluable participants was 62.1% (95% CI, 42.3–79.3).

Immune checkpoint inhibitors (ICIs) should not be used in patients with MCD or KICS because they can exacerbate these conditions. In a phase I study assessing the safety of pembrolizumab in PWH and advanced cancer, a heavily pretreated patient with Kaposi sarcoma and prior history of KICS and elevated peripheral blood mononuclear cell-associated KSHV developed marked KSHV viremia and inflammation, leading to death.[146] Therefore, if the patient has a history of KSHV-associated diseases, ICIs should be used with caution and with consideration of more frequent monitoring for signs and symptoms of KICS or MCD (also see the NCCN Guidelines for Cancer in People with HIV, Principles of Systemic Therapy and Drug-Drug Interactions, available at www.NCCN.org).

Pomalidomide was studied in a phase I/II trial of seven people without HIV and 15 PWH who had Kaposi sarcoma.[72] PWH were required to have viremia controlled and either progressive or stable Kaposi sarcoma on ART. Most of the participants (17 of 22; 77%) had previous therapy for Kaposi sarcoma, exclusive of ART.[147] The response rate was 60% in the PWH group (95% CI, 32%–84%). Grade 3–4 adverse events that might have occurred due to pomalidomide were neutropenia, infection, and edema. Pomalidomide has received accelerated U.S. Food and Drug Administration (FDA) approval for the treatment of adult patients with HIV-related Kaposi sarcoma after highly active ART proves insufficient and for patients without HIV who have Kaposi sarcoma.

Evidence for the use of Albumin-bound Paclitaxel in Kaposi sarcoma appears to be limited to one abstract of a phase II trial of 6 patients with classic Kaposi sarcoma and one case report.[148,149] Partial (n = 2) or complete responses (n = 4) were seen in all patients in the phase II trial.[148] Grade 3 adverse events were neutropenia in half of the patients and thrombocytopenia in 1 of 6 patients. In the case report, Albumin-bound Paclitaxel was effective in a patient with HIV-related Kaposi sarcoma who had relapsed disease after Liposomal Doxorubicin.[149] Furthermore, treatment was associated with an increase in CD4+ T-cell counts.

Bortezomib was studied in the dose-escalation, pilot AMC-063 trial, which included 17 patients with relapsed/refractory HIV-related Kaposi sarcoma on ART.[150] The maximum tolerated dose was not reached. The PR rate was 60% in 15 evaluable patients and 83% in the 1.6 mg/m^2 cohort. The rest of the participants experienced SD. The most common adverse events were diarrhea, fatigue, and nausea.

Etoposide has been studied in multiple phase II trials and in the A5264/AMC-067 trial of patients with HIV-related Kaposi sarcoma.[69,151-153] In one of the phase II trials, 36 patients with previously treated HIV-related Kaposi sarcoma received a course of oral Etoposide, and the overall response rate was 36%, with SD occurring in 33% of the participants.[153] The median duration of response was about 6 months. Grade 3–4 neutropenia occurred in 28%, and opportunistic infections occurred in 22%. The other phase II trials also showed oral Etoposide to have clinical activity and be well tolerated. In the A5264/AMC-067 trial, 190 patients with mild-to-moderate HIV-related Kaposi sarcoma in Africa and South

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America were randomized to ART alone with Etoposide given for progression or ART plus immediate etoposide.[69] No difference in response between the groups was seen at 48 months. If oral Etoposide is used, the Panel recommends the dose escalation used in this trial, as indicated in the guidelines. Etoposide also has activity in classic Kaposi sarcoma.[86,154]

Evidence for the use of Gemcitabine in patients with refractory HIV-related Kaposi sarcoma comes from a retrospective analysis of 23 patients who had been treated with first-line ABV.[155] CR was seen in three patients (13%), PR in eight (35%), and SD in 11 (48%). Only one patient had PD. Grade 3–4 adverse events include leukopenia, pain, fatigue, and neutropenia. Gemcitabine has also been studied as first-line systemic therapy in patients with HIV-related Kaposi sarcoma in a phase IIA trial in West Kenya, with a CR rate of 33% and a PR rate of 53%.[156] Gemcitabine activity has been demonstrated in classic Kaposi sarcoma.[86,157]

Imatinib has activity in HIV-related Kaposi sarcoma.[158,159] The strongest evidence comes from a multicenter phase II trial in which 30 patients were treated with Imatinib.[160] Eighteen patients (60%) had received prior therapy. Partial response occurred in 33%, and 20% had stable disease. The median duration of response was approximately 8 months, with disease progression in seven patients (23%). Grade 3–4 adverse events attributed to imatinib included allergic reaction/hypersensitivity, nausea, dehydration, and cellulitis, but only five patients (17%) discontinued therapy because of adverse events.

Evidence for the use of Ipilimumab + Nivolumab is limited to a phase II single-arm study in 18 patients with progressive disease after one or more lines of systemic therapy.[161] The primary endpoint of overall response rate was 87%. The secondary endpoint of 6-month progression-free survival rate was 76.5%. Grade 3–4 adverse events were experienced by four patients (one patient with grade 3 colitis, two patients with grade 3–4 lipase elevation, and one with grade 3 pneumonitis). Two patients discontinued treatment due to adverse events possibly related to treatment.

The phase II AIDS Malignancy Consortium trial evaluated response rates after Lenalidomide in 42 persons with relapsed/refractory HIV-related Kaposi sarcoma.[73] PRs occurred in 60% of participants. The most common adverse events were neutropenia, fatigue, leukopenia, and diarrhea. The phase II ANRS 154 Lenakap trial evaluated the rate of PR or CR at week 24 after treatment with Lenalidomide in 12 patients with relapsed/refractory HIV-related Kaposi sarcoma.[71] The primary endpoint was the rate of PR or CR by Physical Global Assessment (PGA) criteria. Although none of the 10 patients who were evaluable at 24 weeks met PGA criteria at 24 or 48 weeks, four met ACTG criteria for PR at 48 weeks.

Thalidomide has been studied in HIV-related Kaposi sarcoma in two phase II trials.[162,163] One of these trials included 17 assessable patients with PD.[162] PRs were seen in 47%, and SD was seen in 12%. Time to progression was a median 7.3 months. The most frequently reported side effects were drowsiness in 45% of participants and depression in 35%. Although no prospective trials have used Thalidomide for Kaposi sarcoma-associated IRIS, successful control of steroid-refractory IRIS with Thalidomide has been reported, and thalidomide is an active agent in Kaposi sarcoma.[70] The Panel thus believes that Thalidomide and other immunomodulatory drugs may be a useful option for patients with Kaposi sarcoma and corticosteroid-refractory IRIS.

Evidence for the activity of vinorelbine in HIV-related Kaposi sarcoma comes from a phase II trial of 35 assessable patients with PD.[164] Complete clinical responses were seen in 9%, and PRs were seen in 34%. The median duration of response was about 6 months. Neutropenia was the most frequent dose-limiting toxicity, but other side effects were mild and reversible, and the treatment was generally well tolerated.

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Surveillance of Patients with Kaposi Sarcoma

Patients treated for Kaposi sarcoma who do not require active treatment and who are without signs of progression should be followed periodically for life, with timing based on degree of response to therapy and, if applicable, the presence of immunosuppressive therapy or condition (eg, degree of HIV viremia, immune reconstitution). Surveillance should include history and physical (including complete skin and oral exams and documentation of edema and history of additional immunosuppression such as transplant/glucocorticoids), CBC with differential, and comprehensive metabolic panel. For PWH, surveillance should also include T-cell subsets (CD4+ T-cell count) and HIV viral load, and ART adherence should be assessed. All patients should also be educated on the signs of disease progression and encouraged to promptly report any concerns.

If a change in disease is noted, lesions should be photographed for documentation. If there are signs and symptoms concerning for visceral involvement, progression/refractory disease meriting new therapy, or a notable change in disease burden, various tests and imaging studies are indicated based on clinical circumstances. Testing for opportunistic infections should be conducted if signs of infection or inflammation are present. For patients with advanced cutaneous, oral, visceral, or nodal involvement, FOBT and chest x-ray are recommended. In patients with gastrointestinal symptoms such as nausea, vomiting, diarrhea, abdominal pain, and melena/blood in stool, esophagogastroduodenoscopy, colonoscopy, and CT imaging should be performed. When pulmonary symptoms such as dyspnea and cough are present, bronchoscopy and chest CT should be performed.

It is important to note that KSHV is not eradicated with treatment of Kaposi sarcoma, and the risk of future Kaposi sarcoma persists even after complete remission. Avoidance of iatrogenic immunosuppression as well as optimization and monitoring of immune function and HIV control are important to minimize this risk because disease risk generally decreases with immune reconstitution. However, Kaposi sarcoma can persist, relapse, or present even when CD4+ T-cell counts are within the normal range. Less frequent (every 6–12 months) monitoring may be appropriate for select patients with Kaposi sarcoma that is limited in extent, stable for ≥2 years, and without ongoing immunosuppressive therapy and, for PWH, undetectable HIV viral loads, normal T-cell subsets, and regular follow-up with an HIV specialist.

Summary

Management of Kaposi sarcoma depends on location and extent of disease. PWH with limited cutaneous disease that is asymptomatic and cosmetically acceptable to the patient may be treated with ART alone. Remissions or SD may occur with optimization of immune function and HIV viral suppression alone.

Those with symptomatic and/or cosmetically bothersome limited cutaneous disease should be treated with therapy that is minimally invasive with the least toxicity possible, and with ART, if HIV-positive. Options include a limited number of cycles of systemic therapy, topical treatment, intralesional chemotherapy, radiation, cryotherapy, and local excision.

Preferred initial treatment for patients with significant lymphedema and advanced cutaneous, oral, visceral, or nodal HIV-related Kaposi sarcoma is ART with systemic therapy or systemic therapy alone for those without HIV. Alternatively, a well-designed clinical trial of an agent previously demonstrated to have activity is an appropriate option. For those not eligible for systemic therapy, radiation can be used (with ART for PWH). As lymphedema often complicates Kaposi sarcoma, early involvement of a lymphedema specialist is recommended.

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Surveillance for patients treated for Kaposi sarcoma is important, as disease can recur after an initial CR and in the setting of normal values of T-cell subsets. Persistence of KSHV and emergence of distinct tumor clones can lead to disease progression and relapse. Furthermore, because individual Kaposi sarcoma lesions are often distinct clones as opposed to metastases, treatment of existing disease does not prevent occurrence of new lesions.

For relapsed/refractory disease, any systemic therapy that was tolerated with a durable response can be repeated. Otherwise, alternate systemic therapy should be considered.

Additionally, aggressive KSHV-associated conditions may arise synchronously or asynchronously with Kaposi sarcoma, including PEL, MCD, and KICS. Management of these conditions beyond that indicated for Kaposi sarcoma alone is essential, including the use Kaposi sarcoma-directed systemic therapy plus Rituximab as initial therapy in patients with KICS, and the consideration of anti-IL-6-directed therapy for those with refractory KICS.

Immunosuppression in general and glucocorticoids in any formulation should be avoided in patients with active or prior Kaposi sarcoma, or other KSHV-associated conditions, given the potential to cause significant flares or relapses of Kaposi sarcoma. The use of glucocorticoids should be limited to serious or life-threatening conditions for which these therapies are otherwise indicated (ie, anaphylaxis). Clinicians should proceed with caution if using glucocorticoids and consult an HIV or Kaposi sarcoma specialist. Other therapies associated with flares of Kaposi sarcoma include those suppressing B- and T-cell numbers and/or function such as Rituximab and Cyclosporine, respectively.[48,165] Of note, patients with HIV-related lymphomas who have concurrent Kaposi sarcoma are often able to receive multiagent chemotherapy regimens including glucocorticoids and Rituximab without significant flare of Kaposi sarcoma if the regimen also includes agents active against Kaposi sarcoma such as anthracyclines. Careful monitoring of Kaposi sarcoma should be performed during and following conclusion of therapy for lymphoma or autoimmune conditions; additional Kaposi sarcoma-directed therapy may be required after completion of such regimens associated with prolonged immunosuppression (ie, Rituximab, nucleoside analogues).

Overall, the survival of patients with Kaposi sarcoma has greatly improved over time, and long-term survival can be the goal for many patients. The goals of therapy for patients with advanced disease are namely reducing or reversing symptoms and mitigating end-organ damage. Complete remissions in this setting are rare, but effective therapy can result in long-term disease control.

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